

EE 582 – Advanced Topics

(Energy Systems Modeling and Simulation)

Instructor:

Dr. Seyyedmilad (Milad) Ebrahimi

Class Webpage:

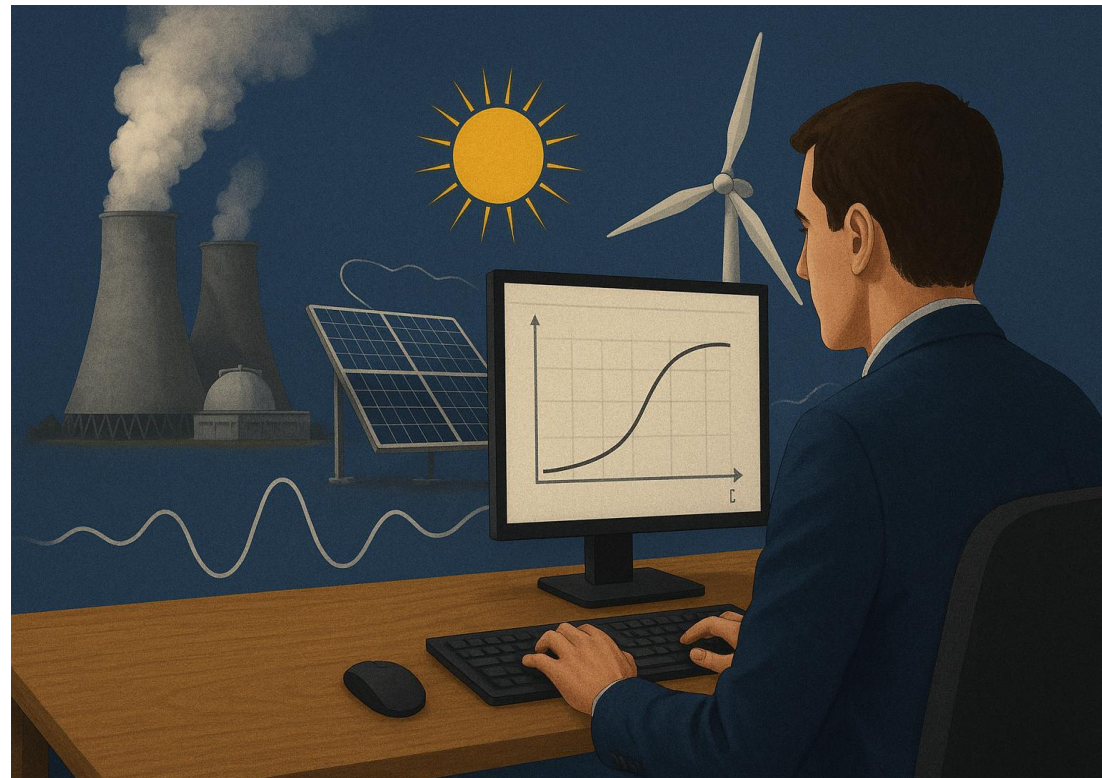
WSU Canvas

Lectures:

Tuesdays / Thursdays

4:20-5:35 pm

Fall 2025

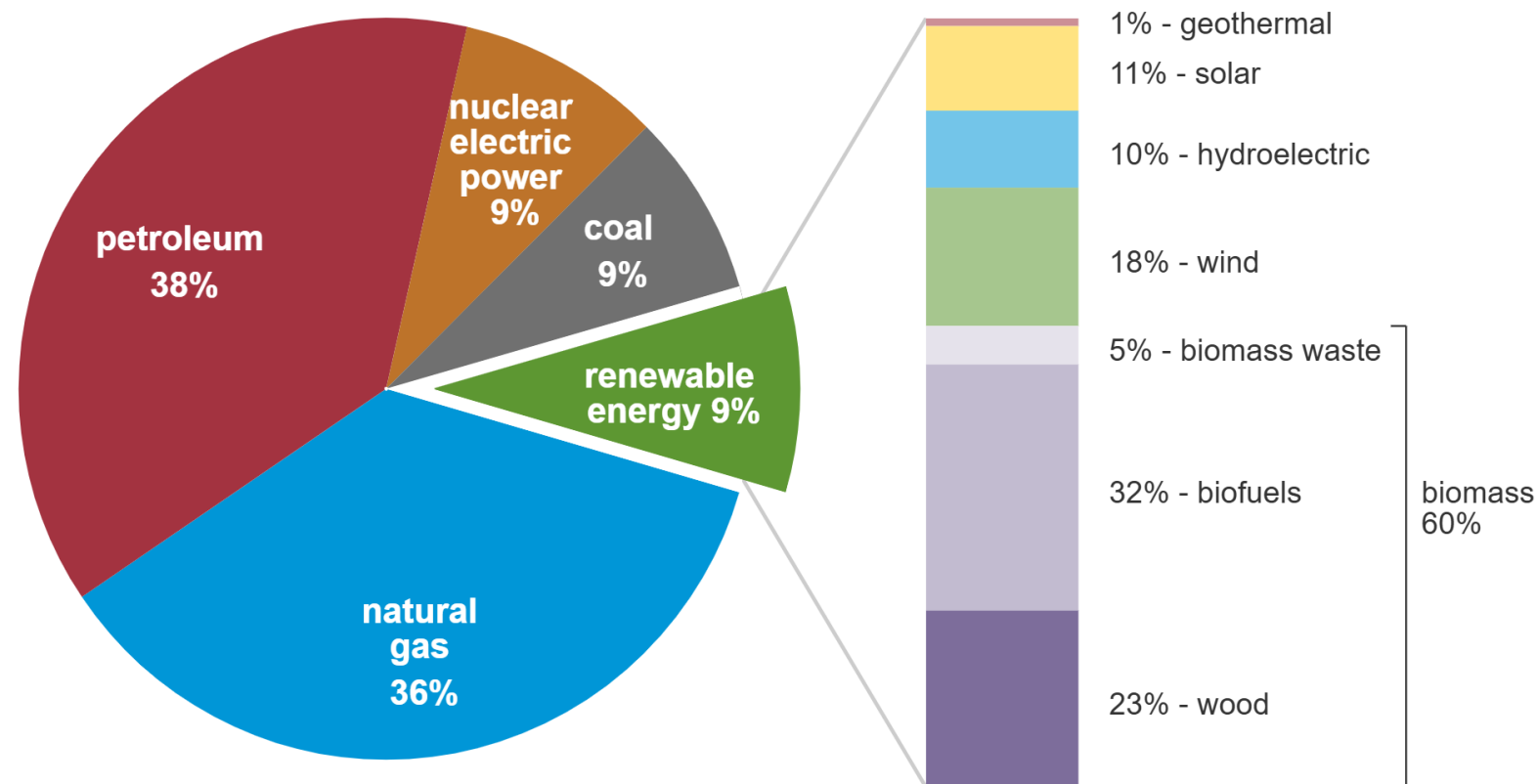


Modern Energy Networks

U.S. primary energy consumption by energy source, 2023

total = 93.59 quadrillion
British thermal units

total = 8.24 quadrillion British thermal units



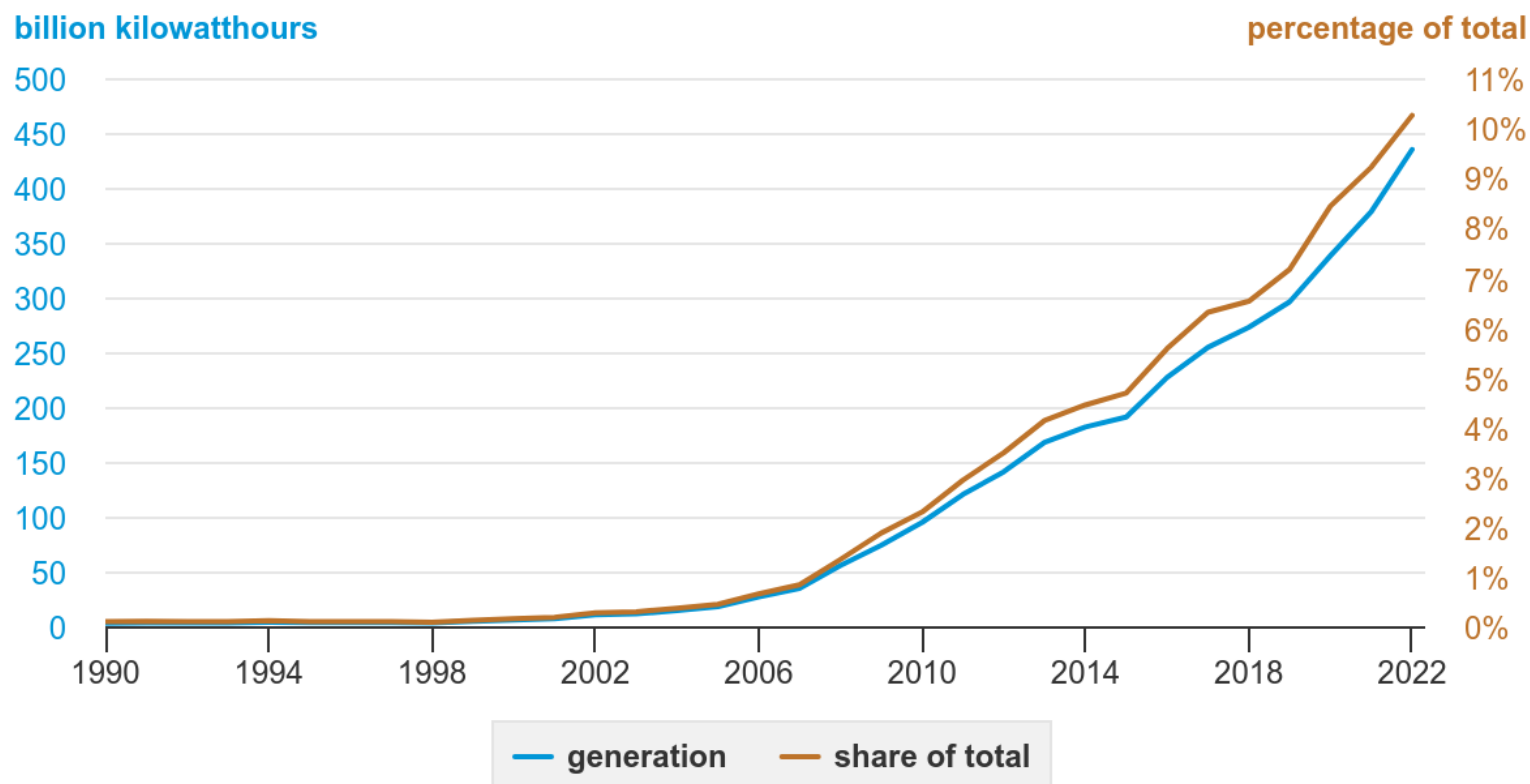
Data source: U.S. Energy Information Administration, *Monthly Energy Review*, Table 1.3 and 10.1, April 2024, preliminary data

Note: Sum of components may not equal 100% because of independent rounding.

Modern Energy Networks

- ✓ Generation of electricity from Wind and Solar power is growing fast

Wind electricity generation and share of total U.S. electricity generation, 1990-2022

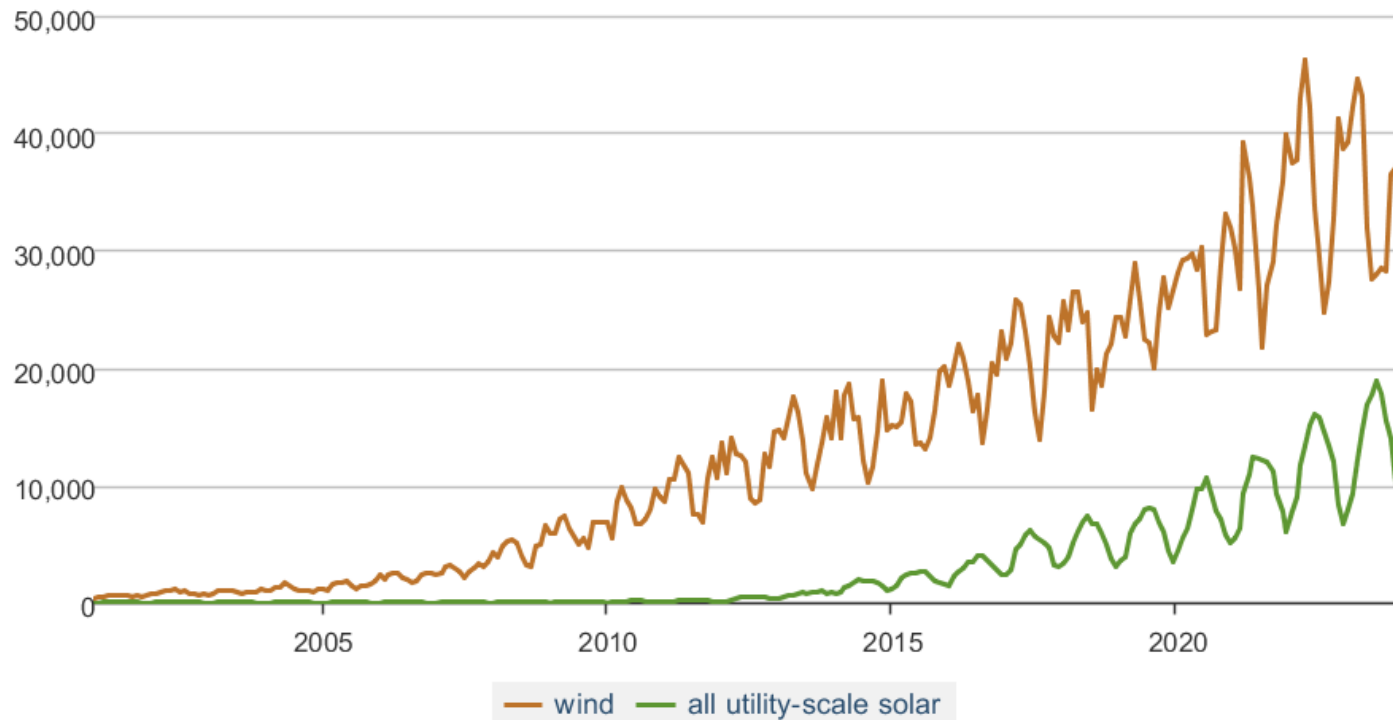


Modern Energy Networks

- ✓ Generation of electricity from Wind and Solar power is growing fast

Net generation, United States, all sectors, monthly

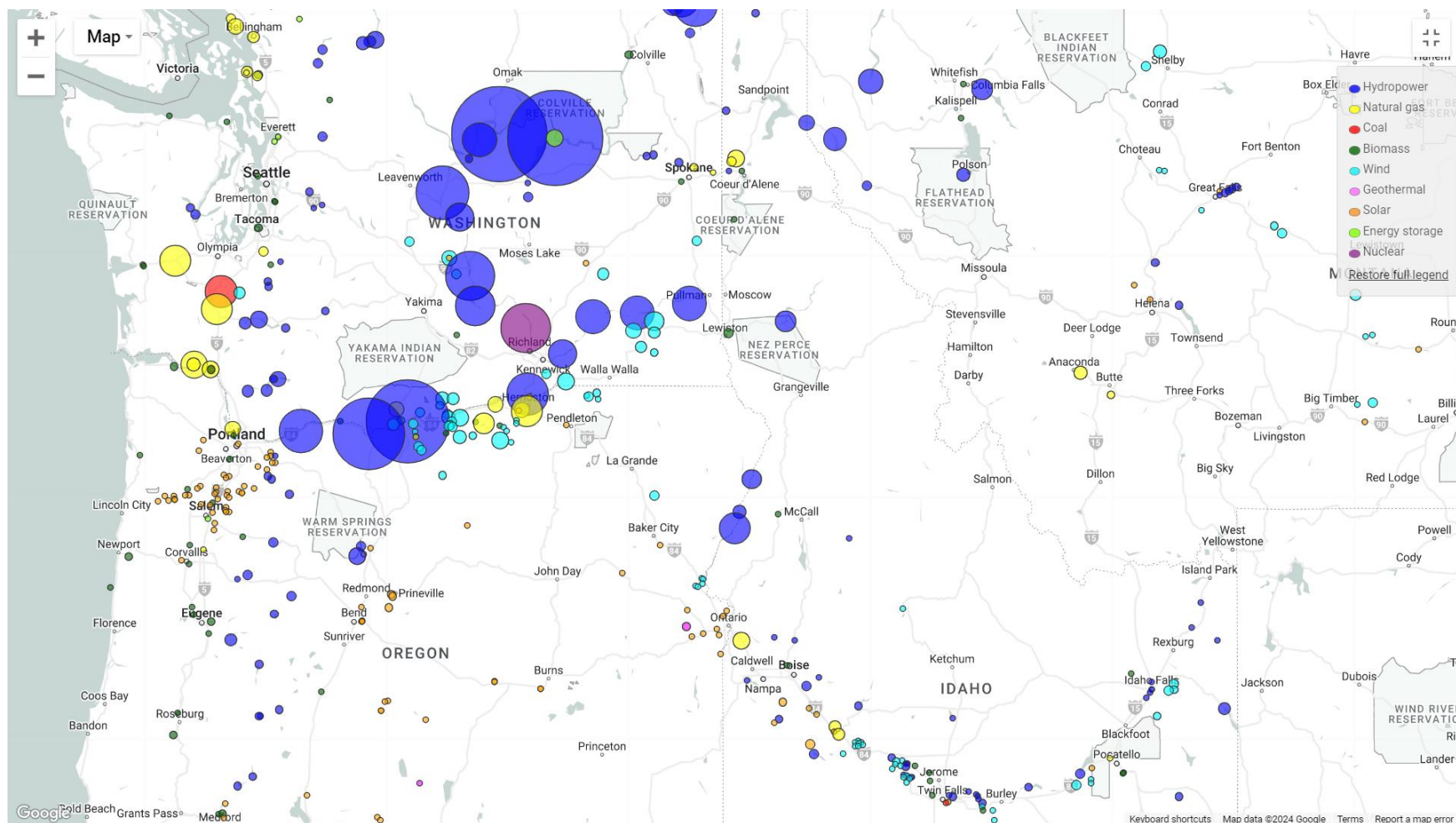
thousand megawatthours



Data source: U.S. Energy Information Administration

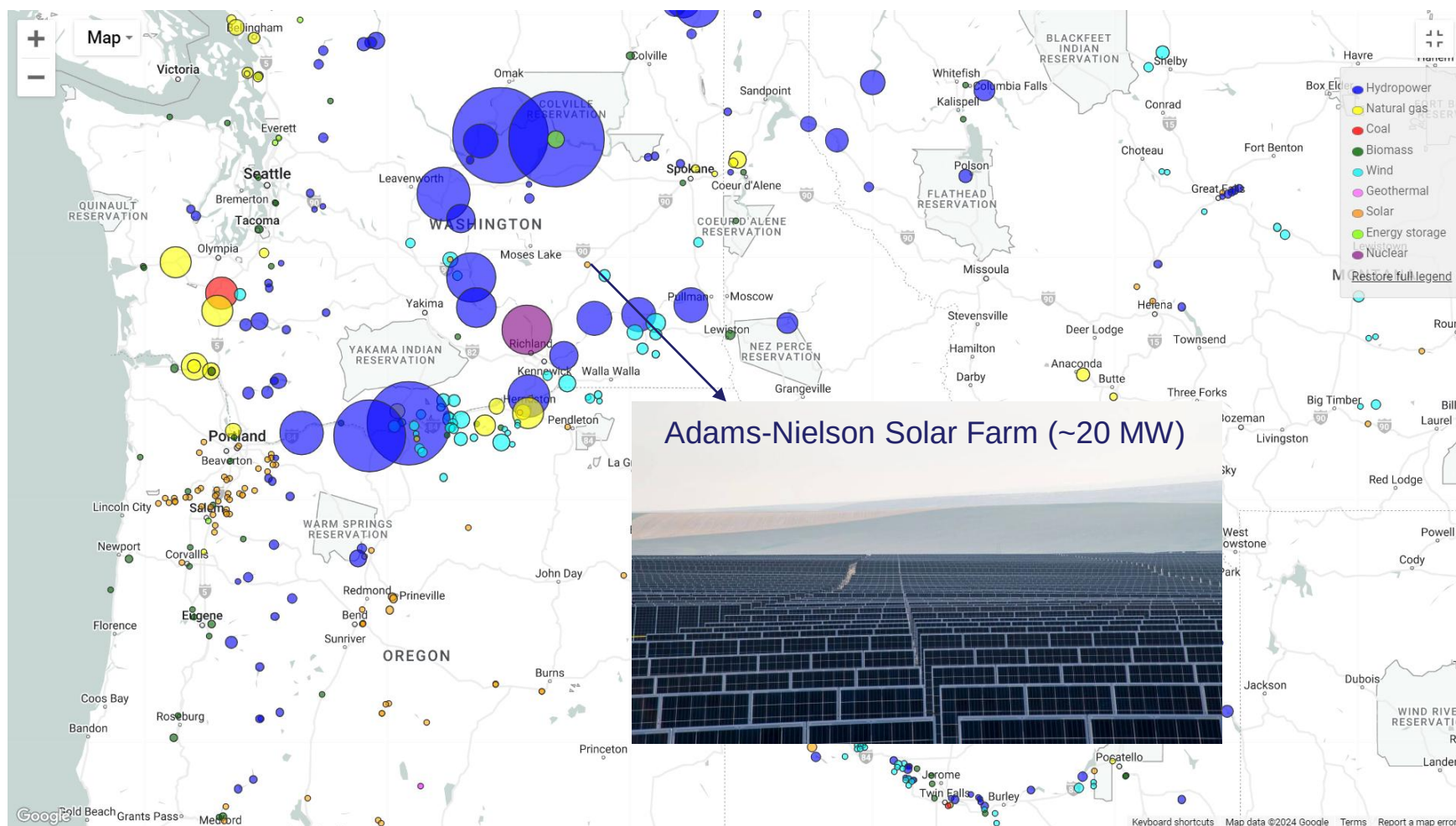
Modern Energy Networks

- ✓ In Northwest, Hydro electric power is the dominant source of renewable energy
- ✓ Generation of electricity from Wind and Solar power is growing fast



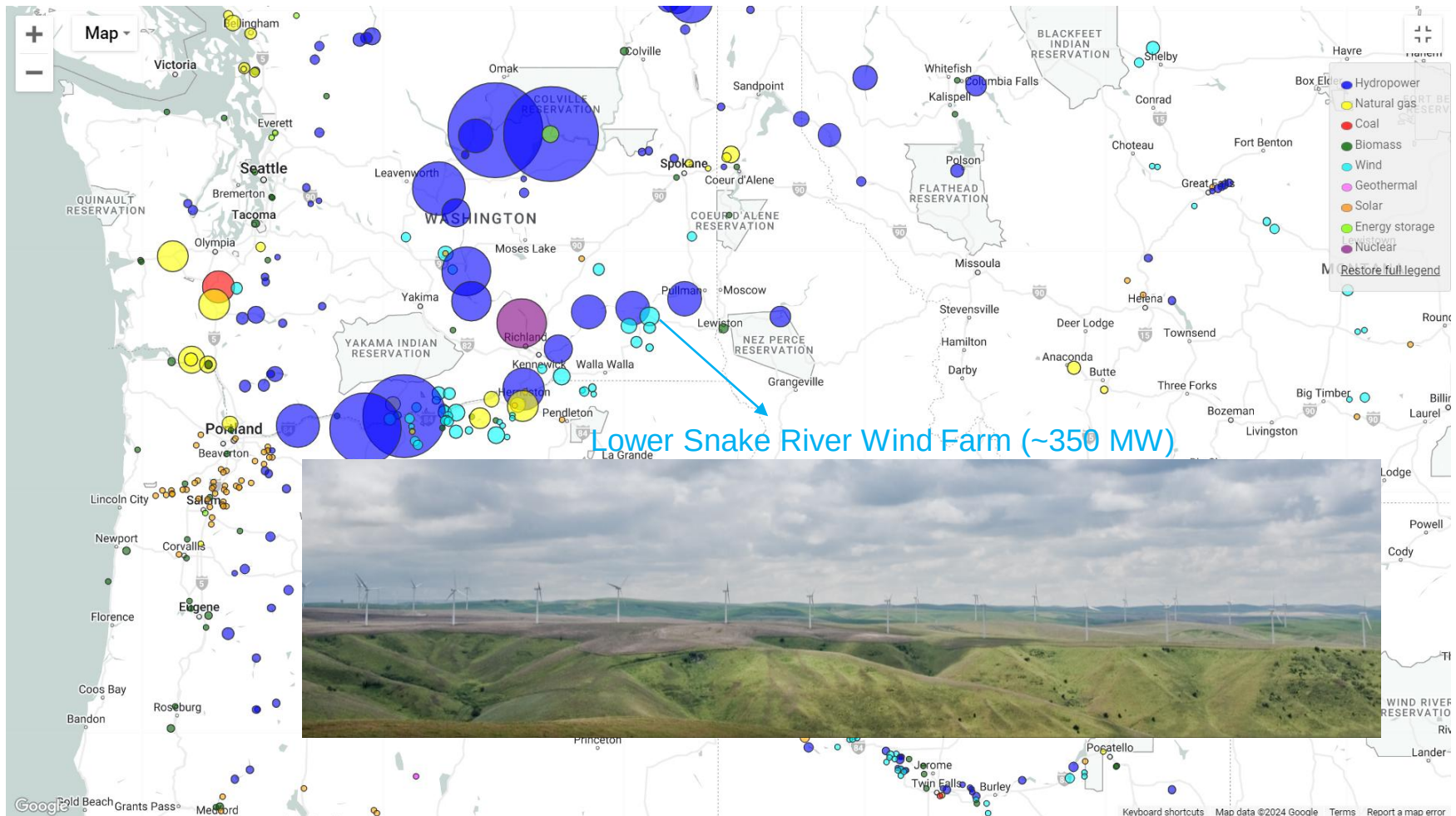
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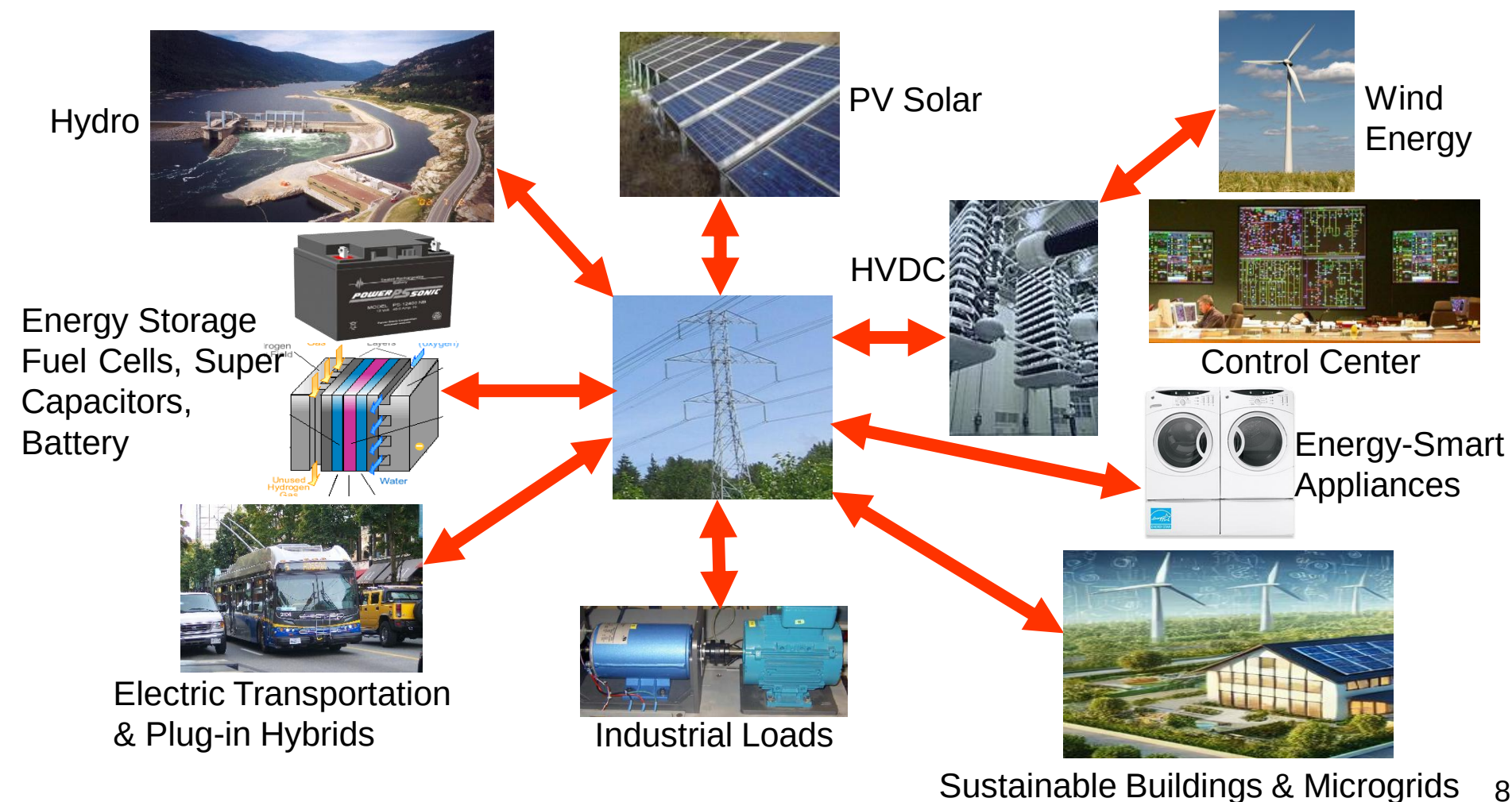
Modern Energy Networks

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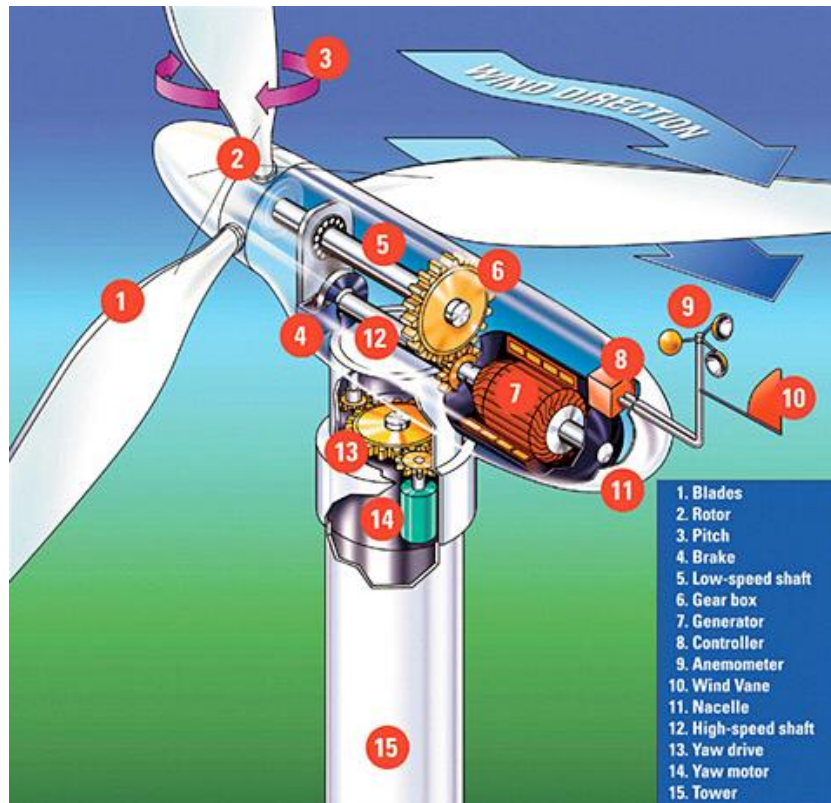
Modern Energy Networks

□ Electric Power & Energy Systems Integration with Power Electronic Converters



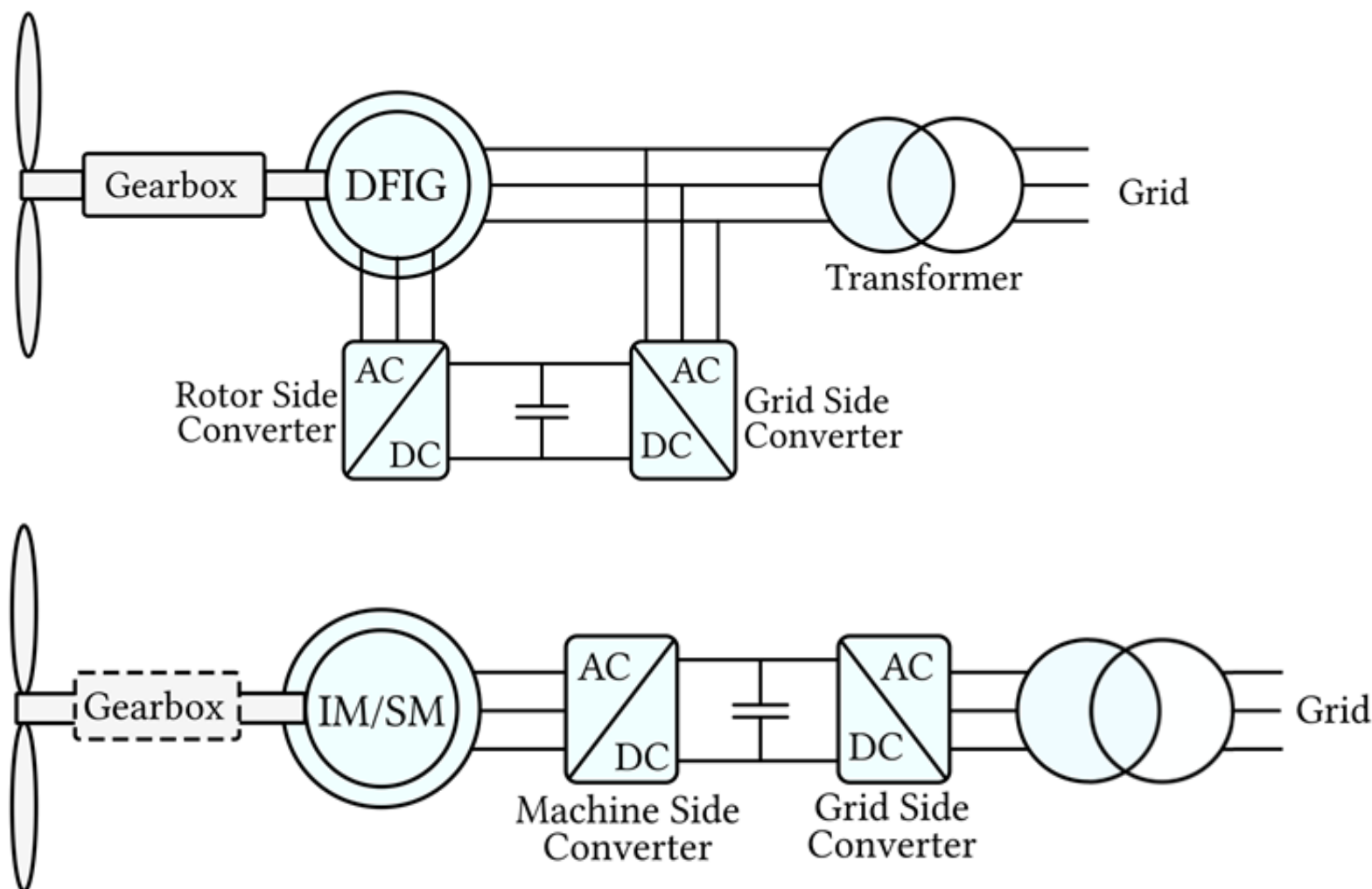
Applications of Power Electronic Converters

❑ Wind Power Generation Systems:



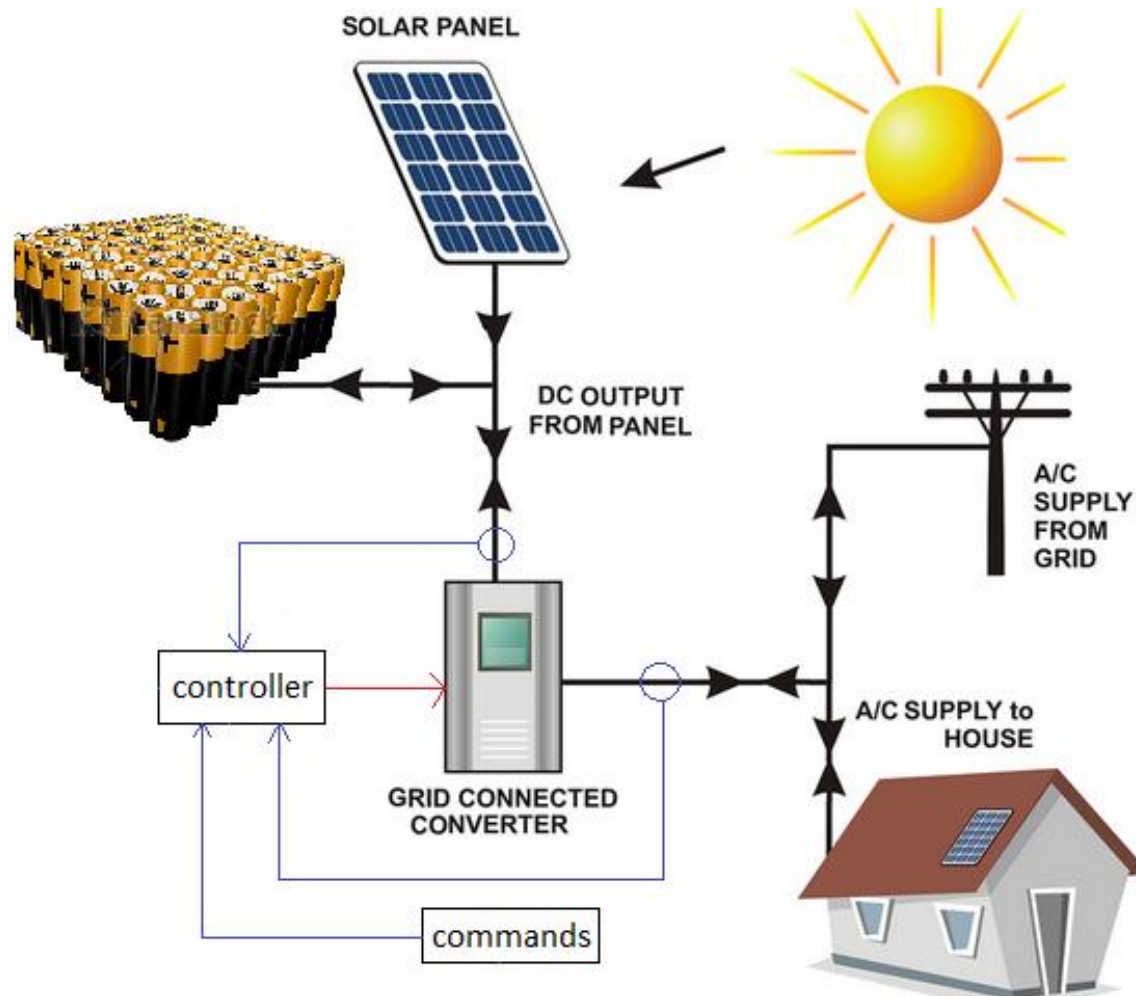
Applications of Power Electronic Converters

❑ Wind Turbine-Generators (Types 3 & 4):



Applications of Power Electronic Converters

☐ Solar Power Generation Systems:

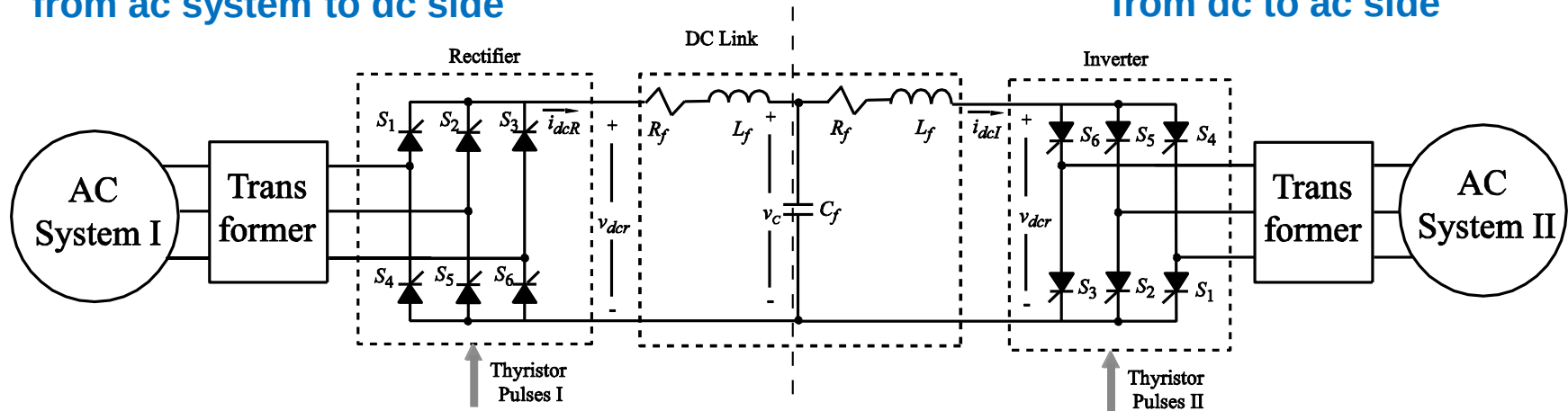


Applications of Power Electronic Converters

□ High-Voltage Direct Current (HVDC) Transmission:

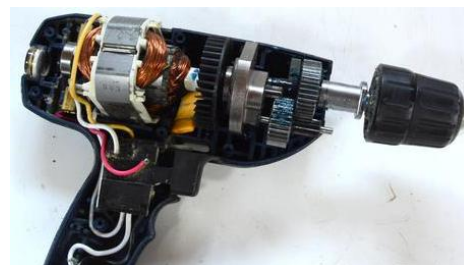
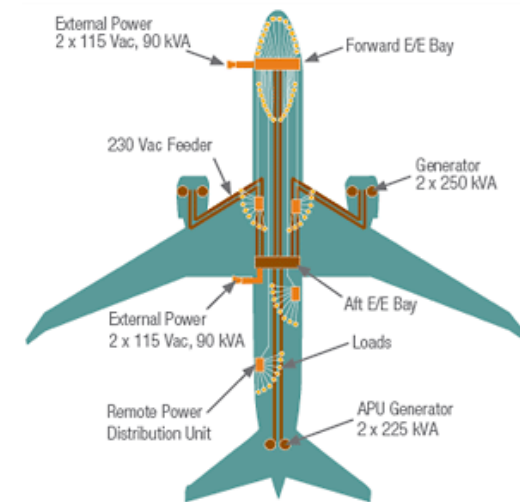
Energy is transferred
from ac system to dc side

Energy is transferred
from dc to ac side



Applications of Power Electronic Converters

❑ Modern Energy Systems:



Applications of Power Electronic Converters

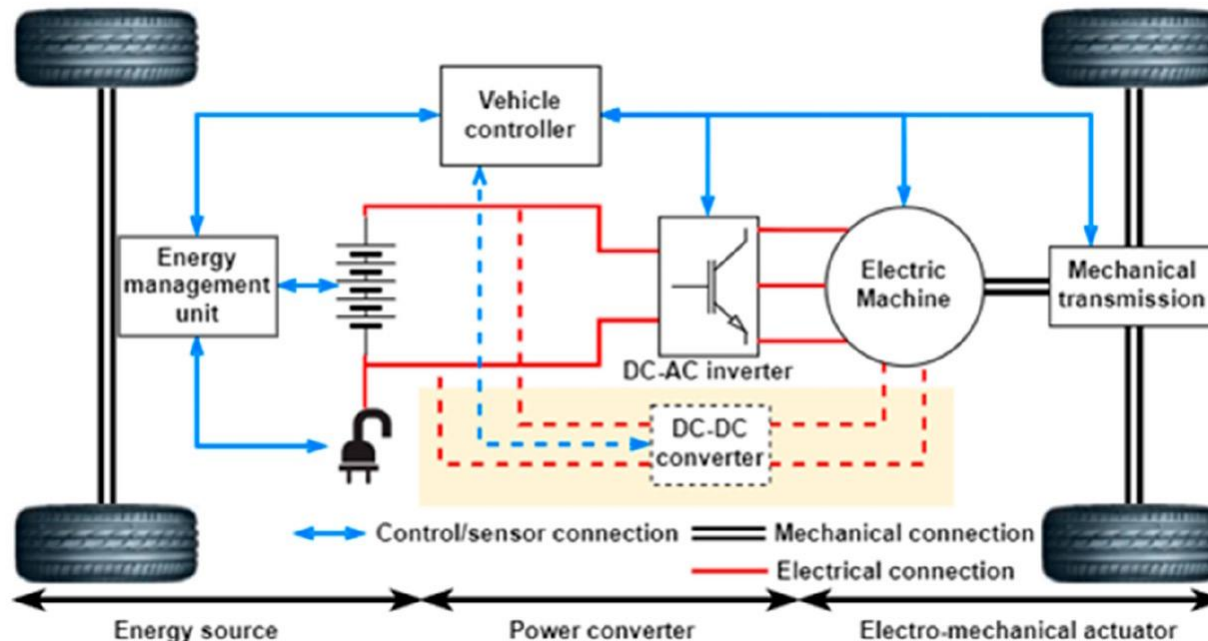
❖ Modern Transportation → More-Electric Vehicles

Three main components of an EV

✓ **Battery**

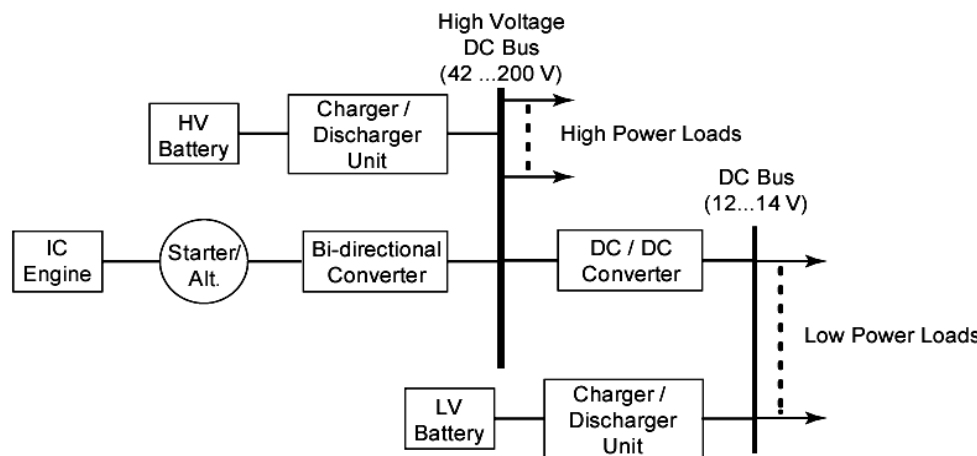
✓ **Electric machine**

✓ **Power-electronic converter**



Applications of Power Electronic Converters

❖ **Modern Transportation** ➡ **More-Electric Vehicles**



Diesel-Electric



Liebherr T282B earth-hauling truck 2.7MW AC Propulsion

Hybrid



Toyota Hybrid, operates at 288V, reaches 30kW

All-Electric ➡ **Zero-Emission Transportation**



Canada Line (Richmond-Airport-Vancouver Line) SNC-Lavalin & Rotem Company



Vancouver TransLink Trolley Bus New Flyer Industries



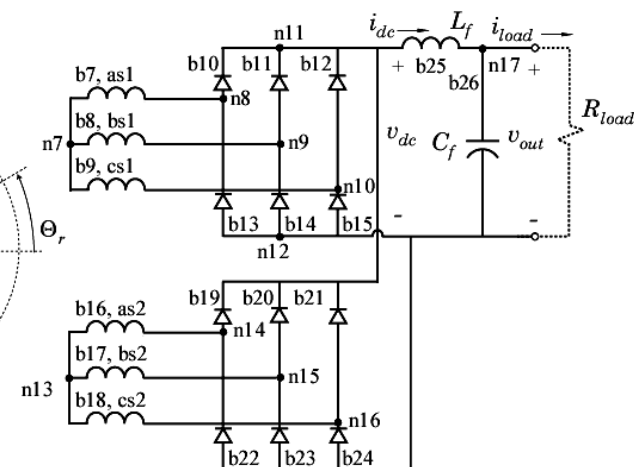
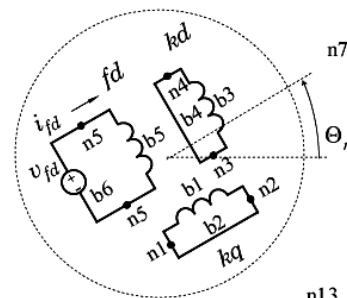
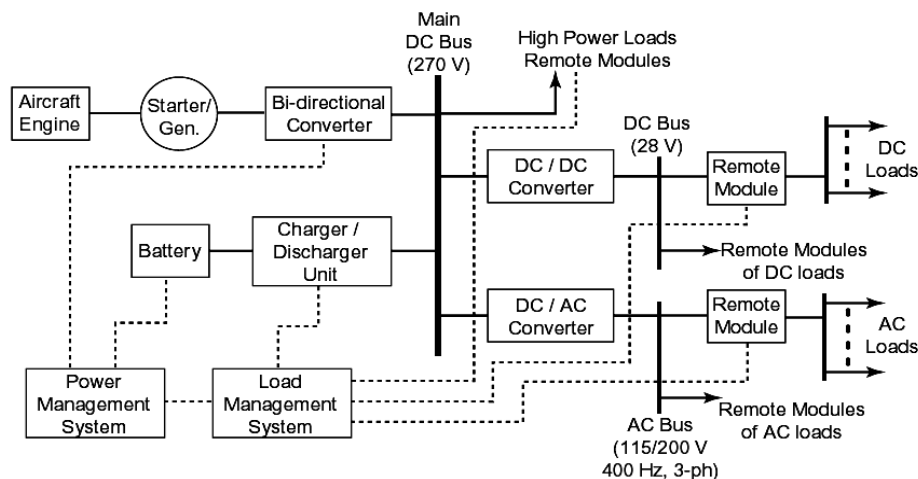
Tesla Roadster, Induction Motor, reaches 200 kW

All-Electric

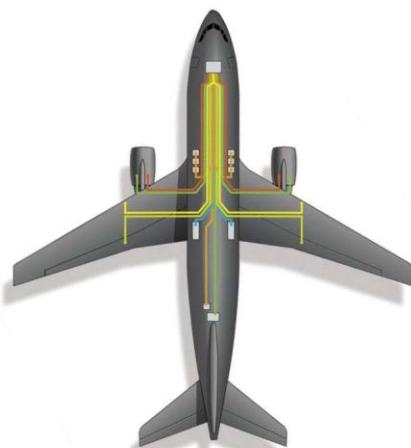


Applications of Power Electronic Converters

❖ Modern More-Electric Aircraft



High-Speed, Low-Weight, High-Phase-Count Motors, Generators and Converters



Airbus A380



**Antares 20E
42kW BLDC Propulsion**



**Electric Toys
BLDC Propulsion**

Applications of Power Electronic Converters

❖ Modern **Electrified Ship** Propulsion Systems

- ✓ utilize renewable green energy resources



solar-powered vessel



wind-assisted vessel

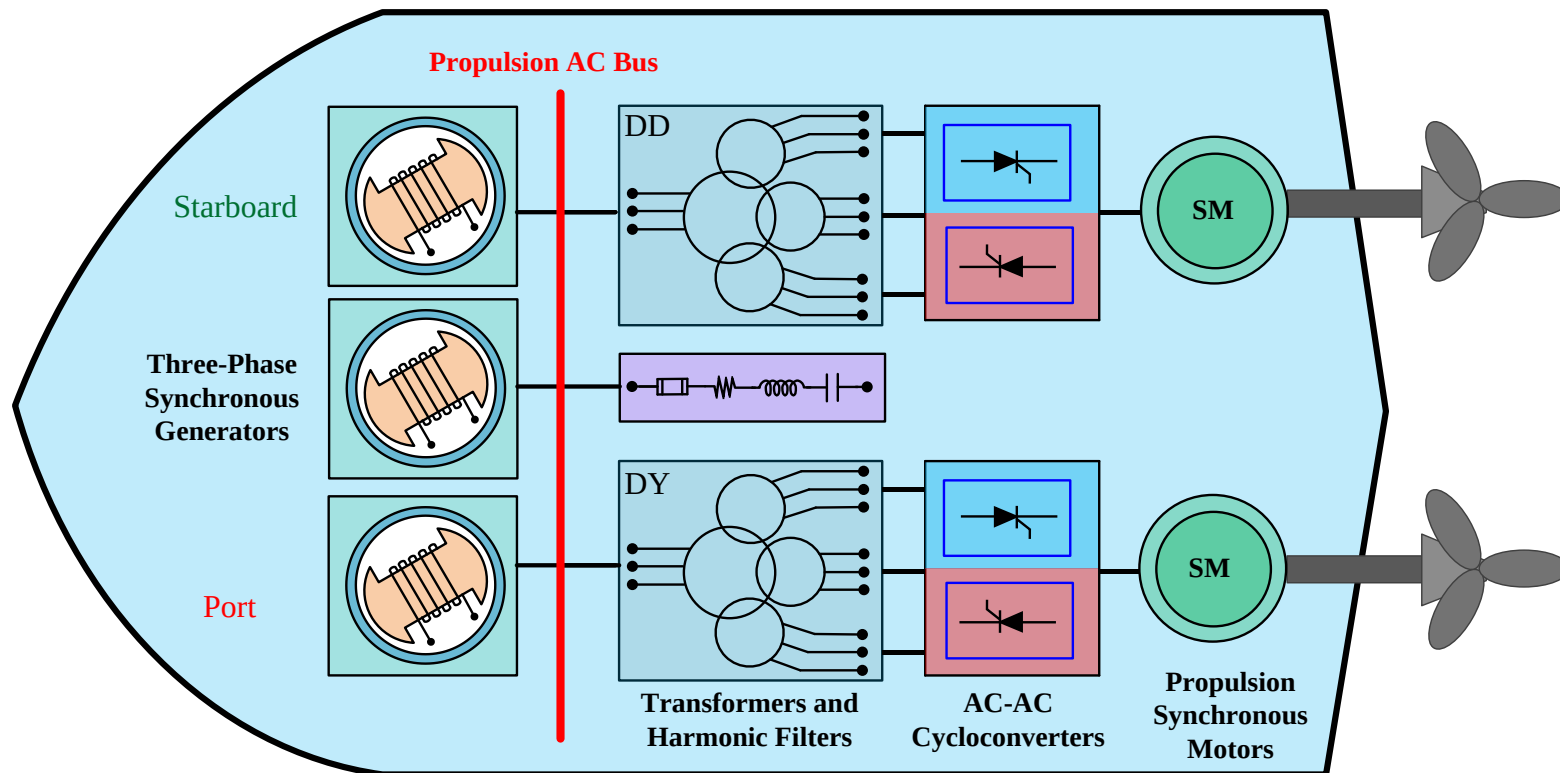
- ❖ There has been a growing trend towards electrifying the propulsion systems of ships



- ✓ increases efficiency
- ✓ reduces fuel consumption and vibrations
- ✓ achieves better overall dynamic performance

Applications of Power Electronic Converters

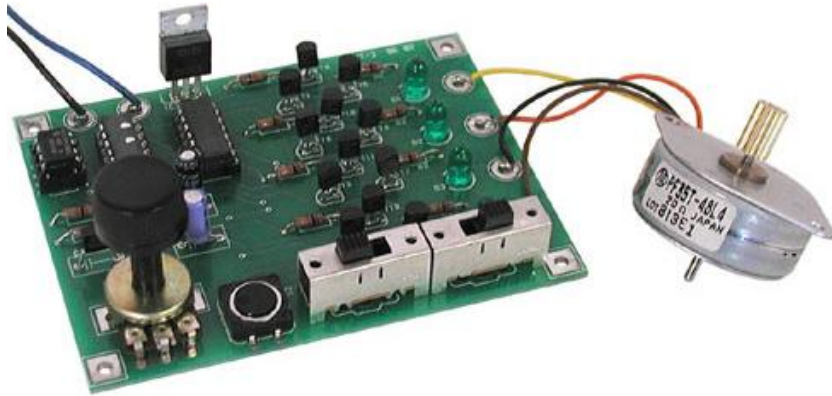
✓ **Modern ships are undergoing electrification**



- **Electric power is provided by multiple synchronous generators**
- **Propulsion system is driven by electric (synchronous) motors**
- **Electric motors are controlled/driven by variable frequency drives (VFDs)**

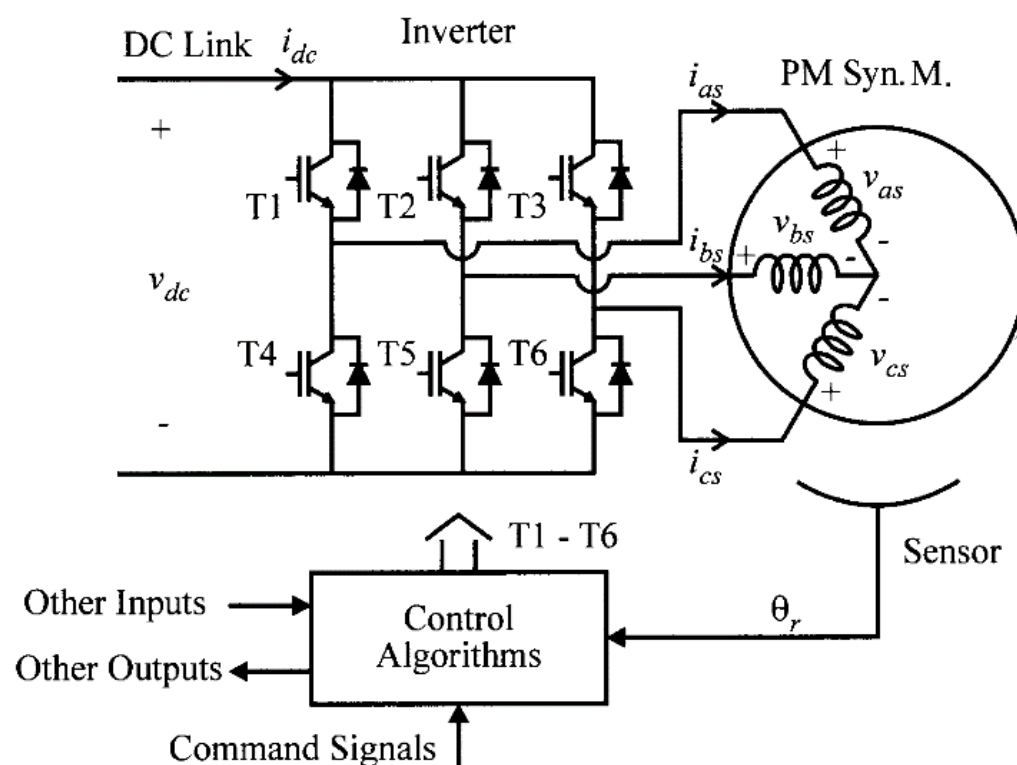
Applications of Power Electronic Converters

❑ Variable Frequency Drives (VFDs):



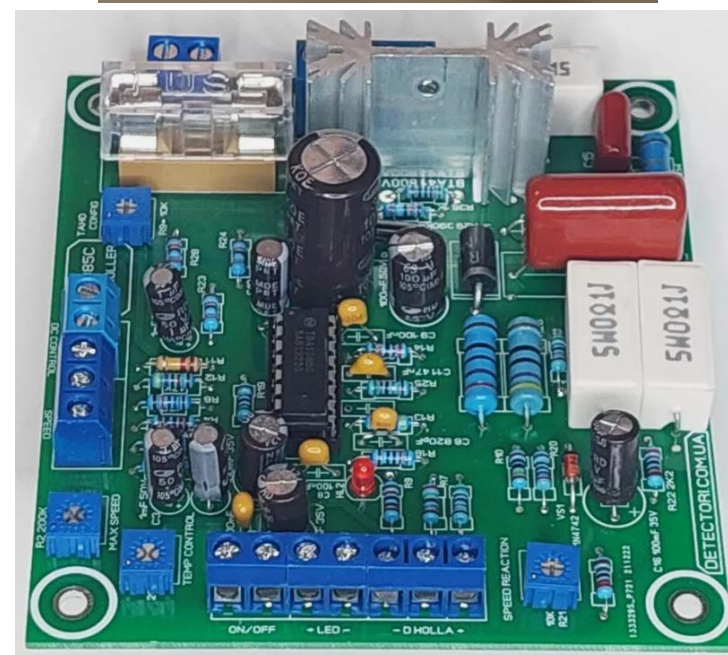
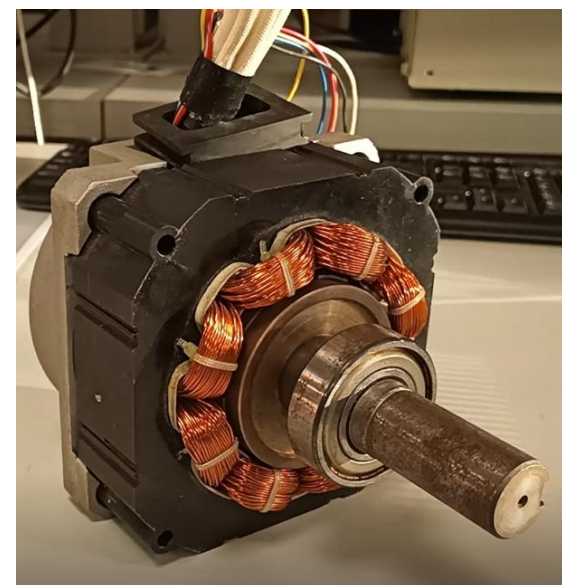
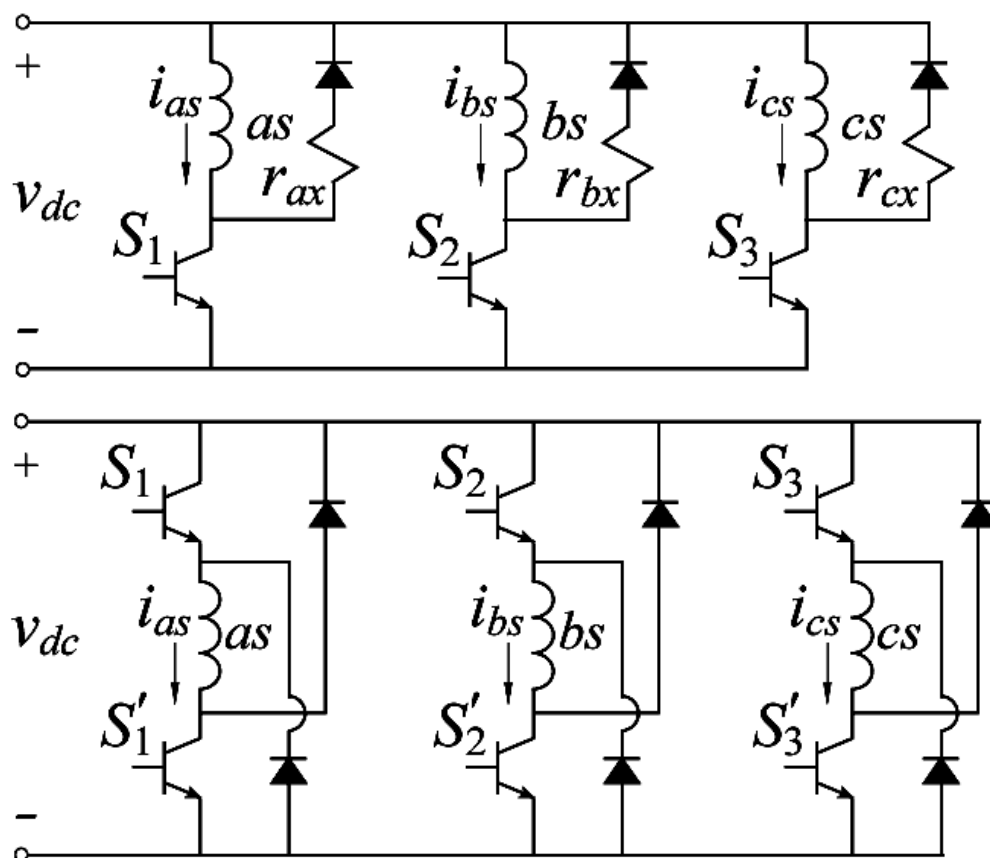
Applications of Power Electronic Converters

❑ Brushless AC or DC Servo Motor Drives:



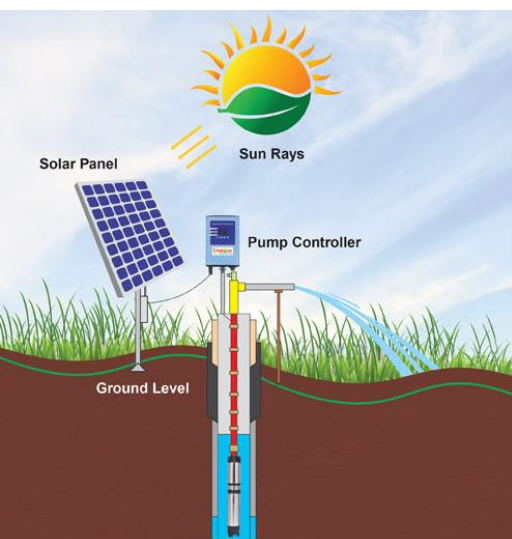
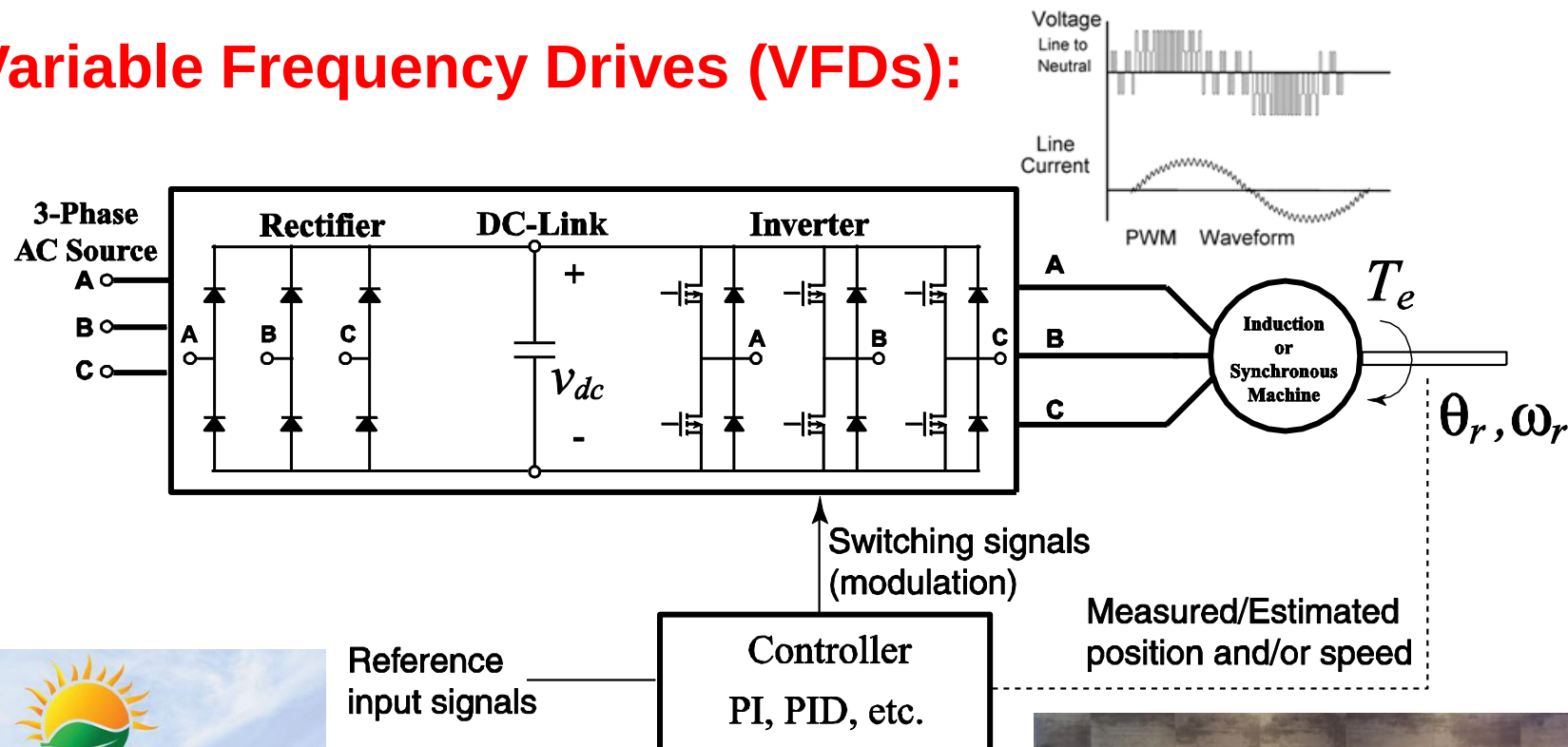
Applications of Power Electronic Converters

❑ Drivers of Variable Reluctance Motors:



Applications of Power Electronic Converters

Variable Frequency Drives (VFDs):

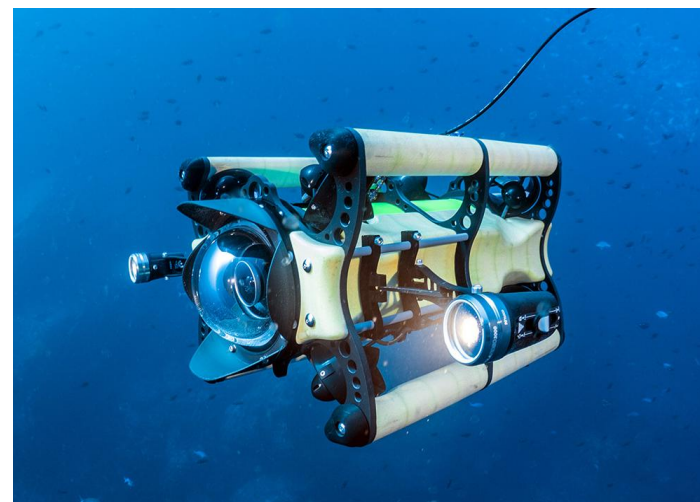
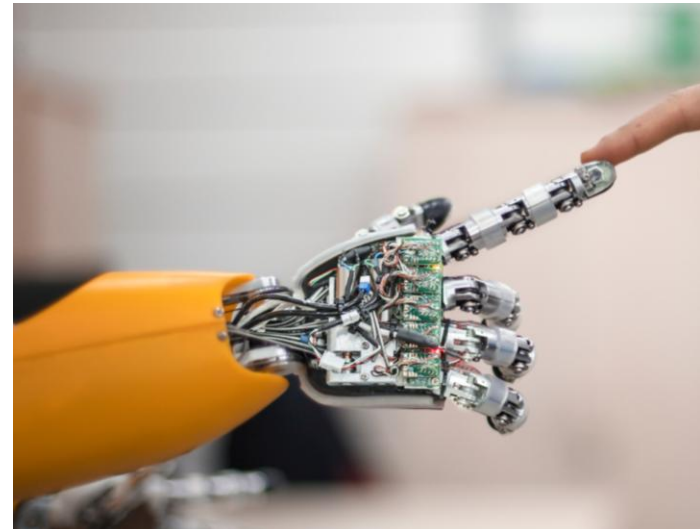


- ✓ Use inverter to modulate AC output with variable frequency and magnitude



Applications of Power Electronic Converters

❑ Industrial Automation, Robots:



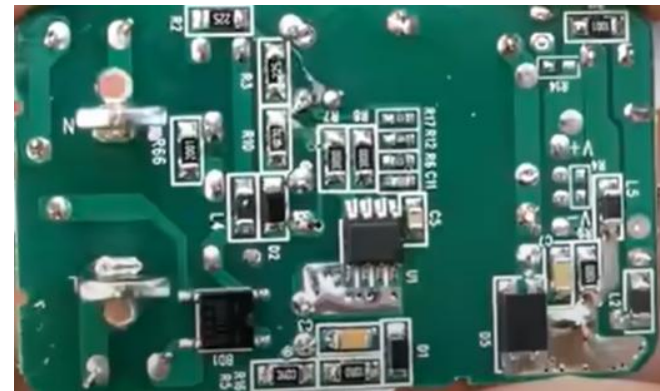
Applications of Power Electronic Converters

❑ Computer Power Supplies:



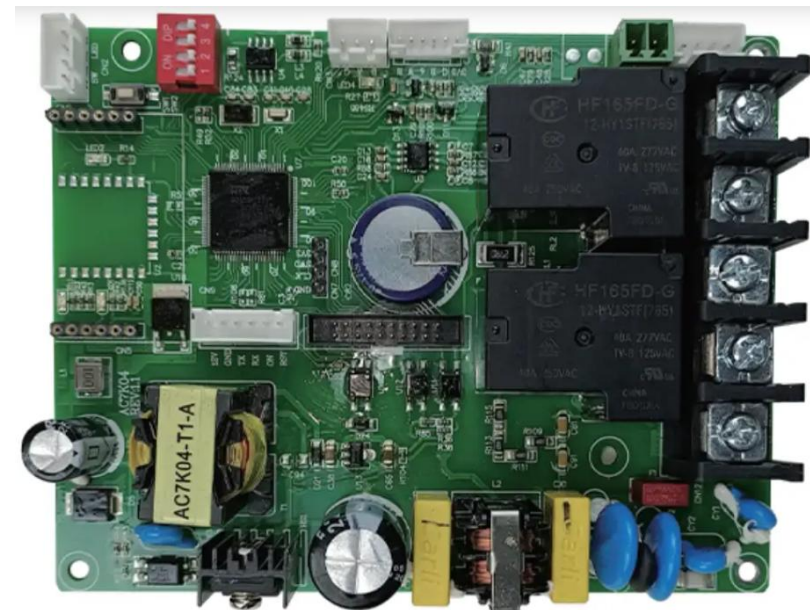
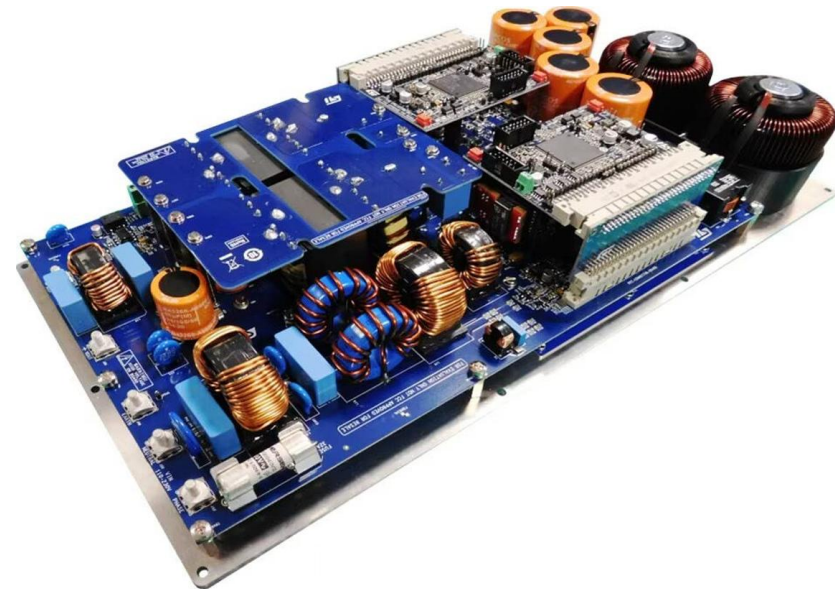
Applications of Power Electronic Converters

❑ Adaptors/Chargers for Laptops and Cellphones:



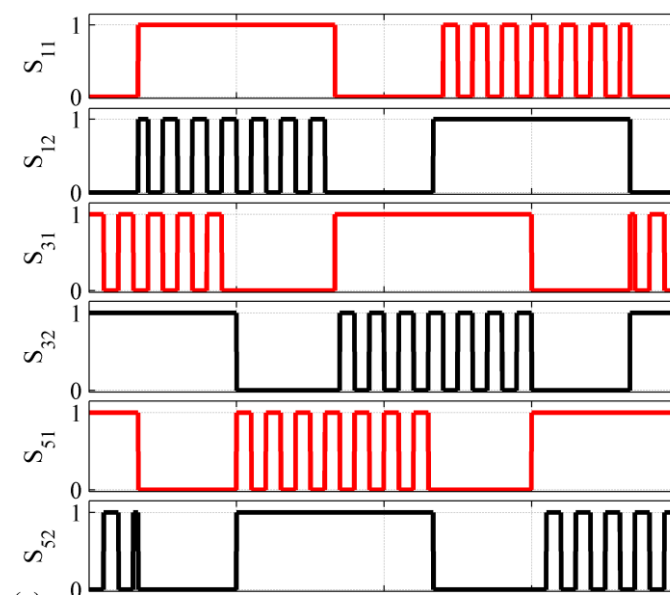
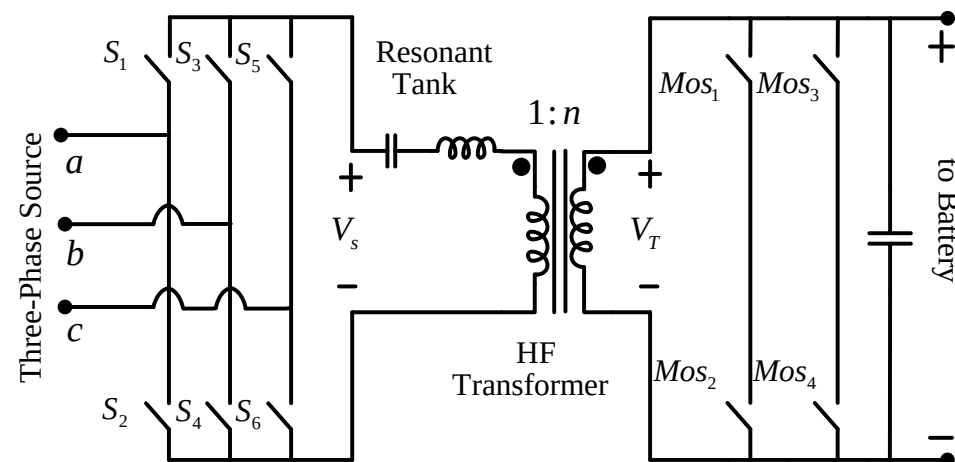
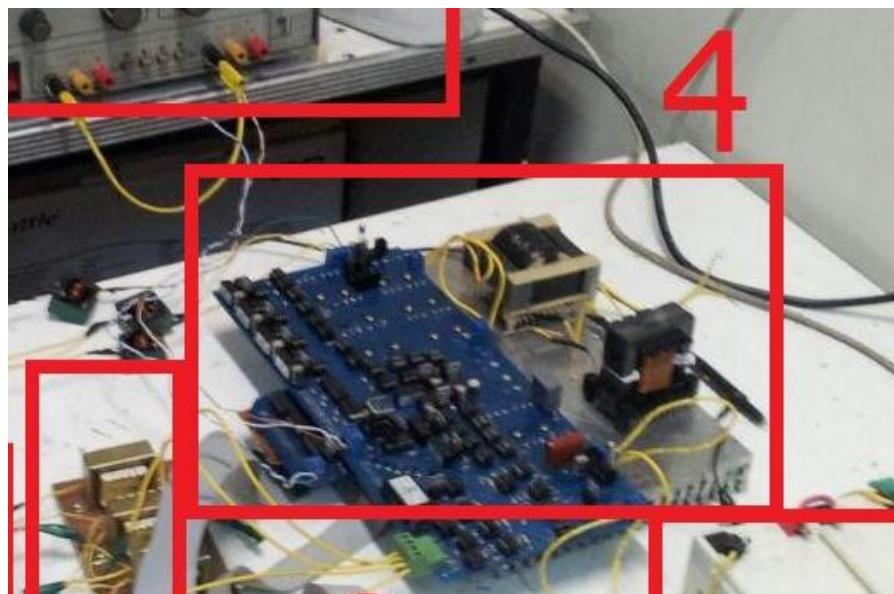
Applications of Power Electronic Converters

❑ Electric Vehicle Chargers:



Applications of Power Electronic Converters

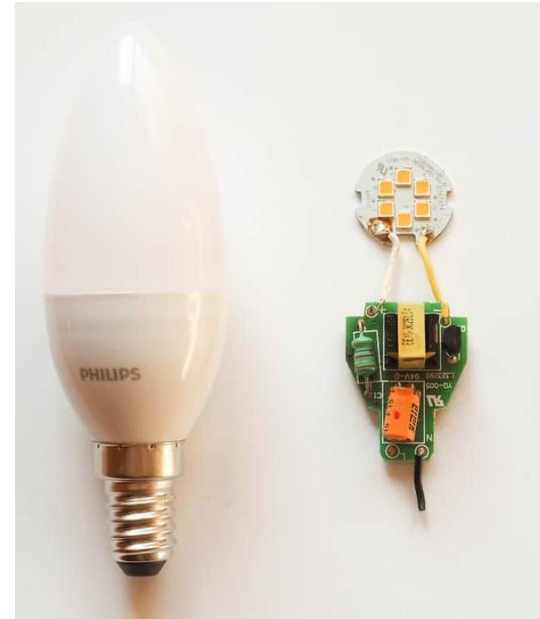
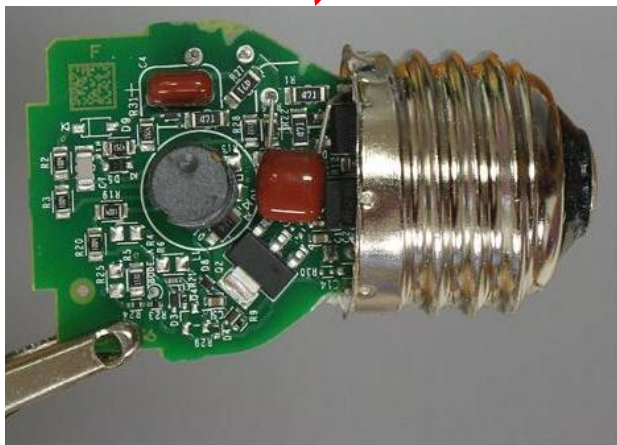
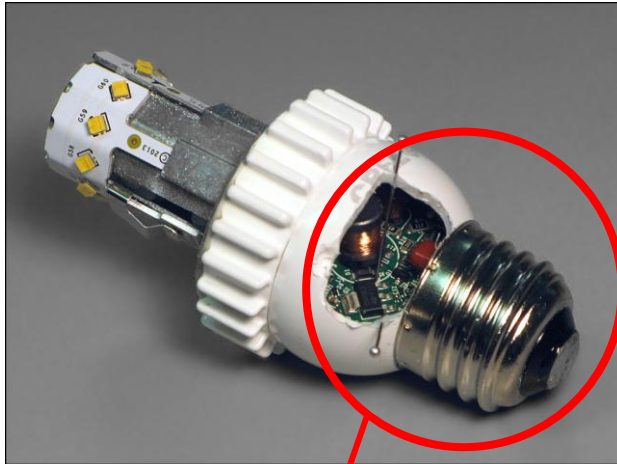
❑ Electric Vehicle Chargers:



- ✓ S. Ebrahimi et. al., “An isolated bidirectional integrated PHEV battery charger with resonant converters,” in *Journal of Electric Power Components and Systems*, Taylor & Francis Group, vol. 44, no.12, pp. 1371–1383, Jun. 2016.

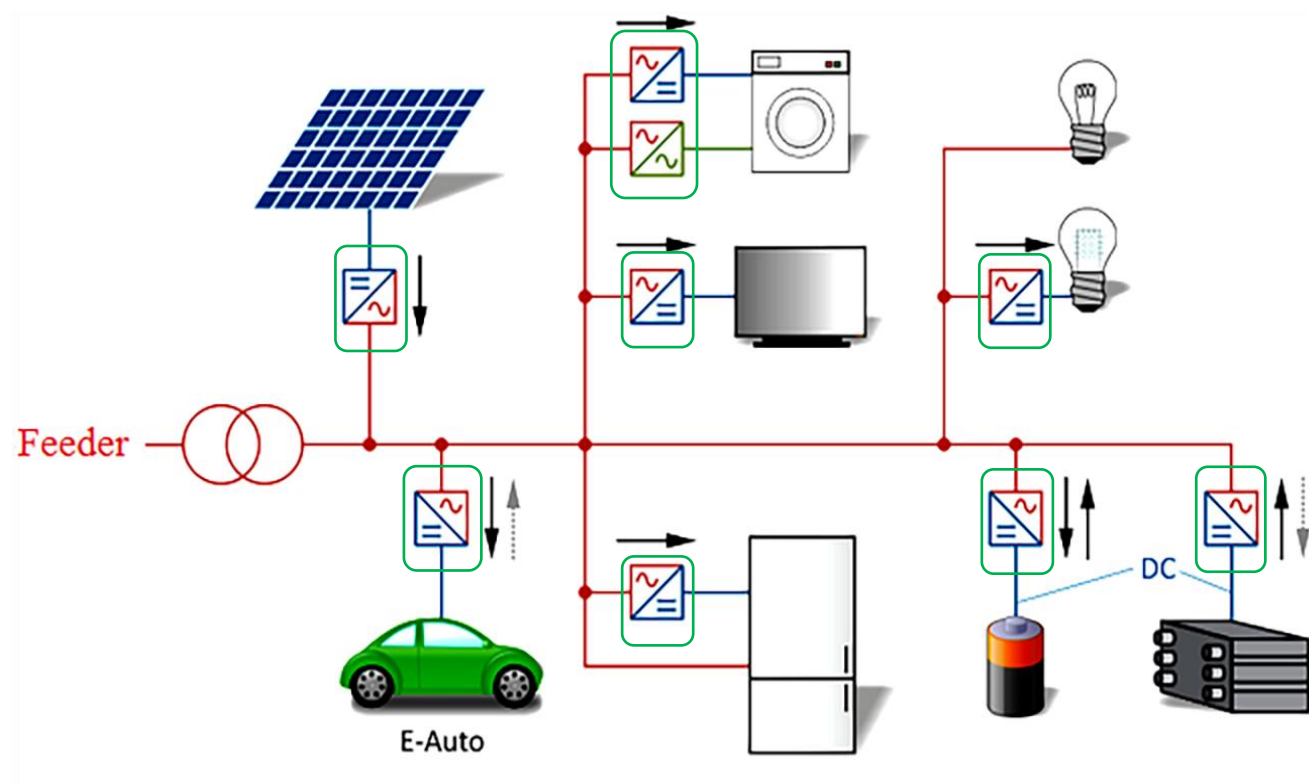
Applications of Power Electronic Converters

❑ Modern Lighting Systems:



Applications of Power Electronic Converters

❑ Modern Energy Distribution Systems:

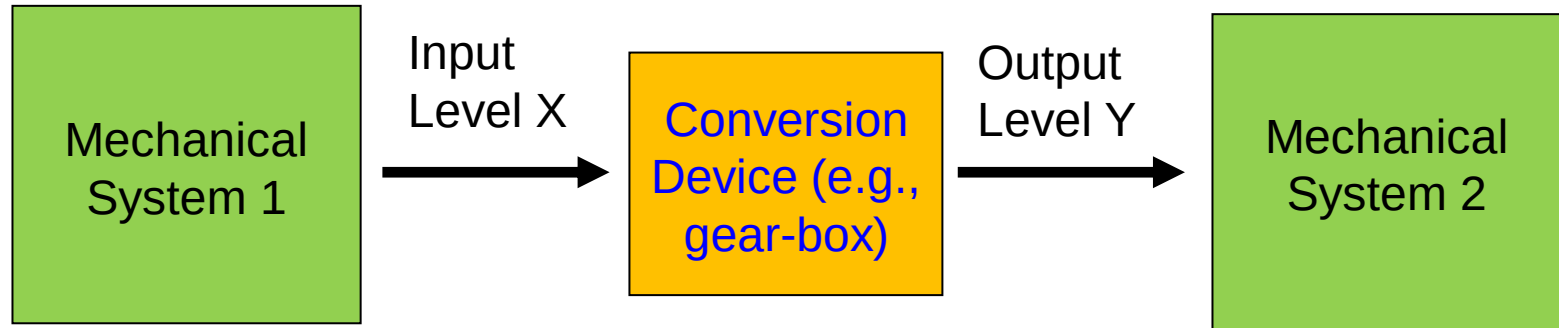


➤ Include many power-electronic converters

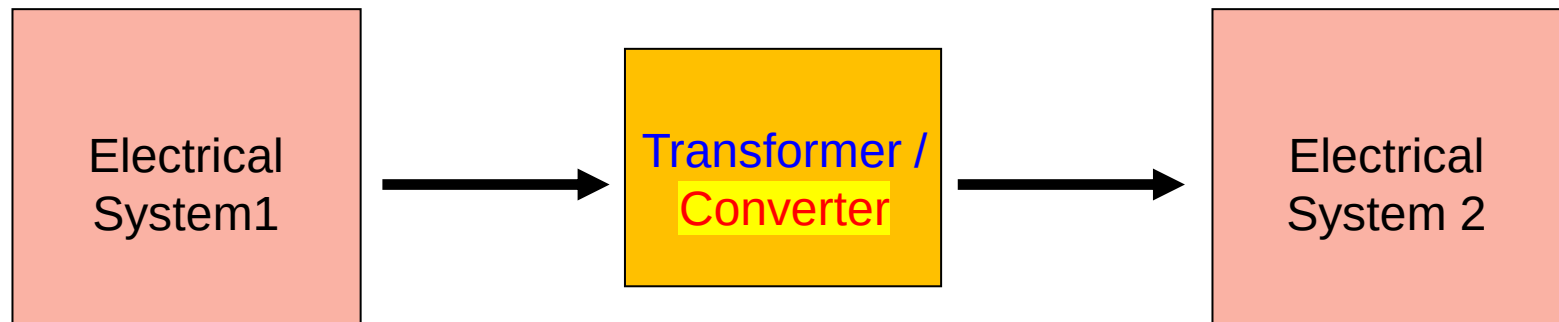
By 2030, 80% of electricity could flow through power electronic devices
[U.S. Department of Energy Office of Scientific and Technical Information]

Energy Conversion Systems

- Transformation of Energy (Mechanical to Mechanical)

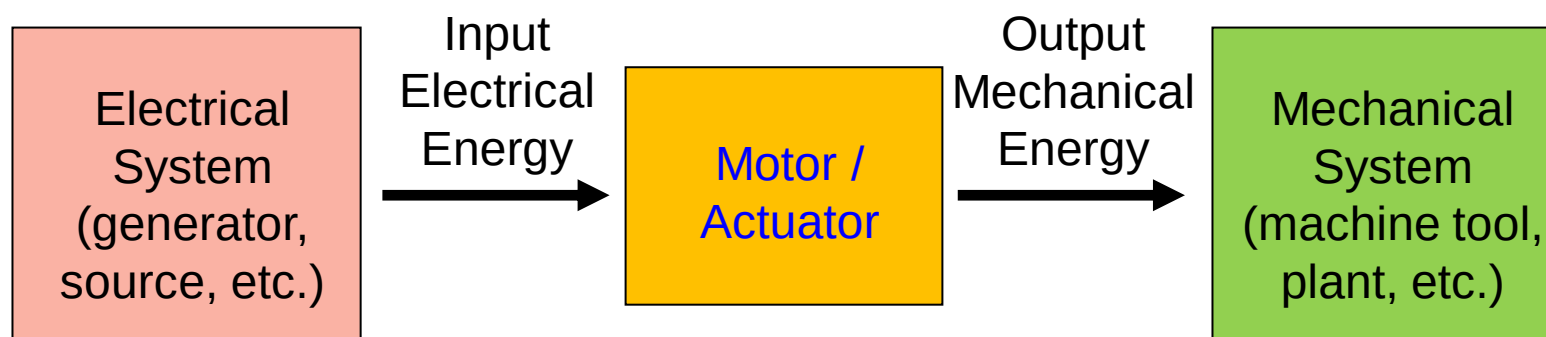


- Transformation of Energy (Electrical to Electrical)

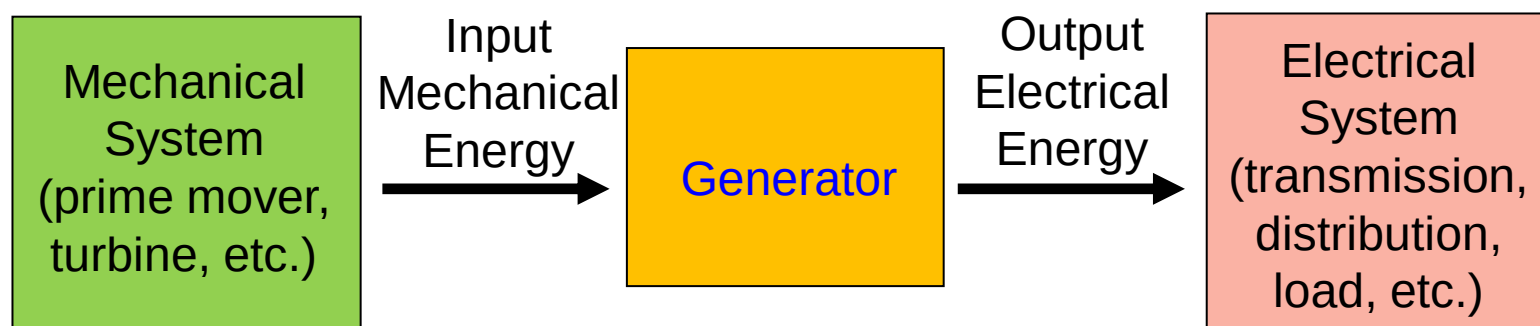


Energy Conversion Systems

- Transformation of Energy (Electrical to Mechanical)

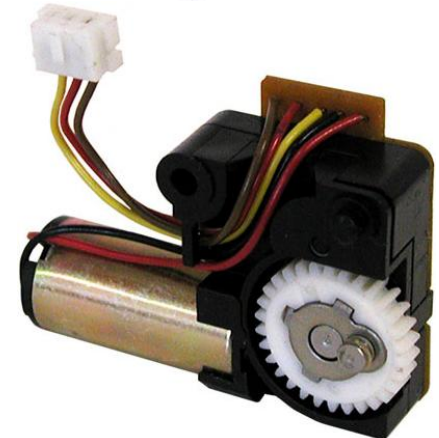


- Transformation of Energy (Mechanical to Electrical)



Electromechanical Devices

- ✓ Manufacturing
- ✓ Automotive
- ✓ Aircraft
- ✓ Ships
- ✓ Computers
- ✓ Office
- ✓ Household
- ✓ ...



Generation of Electricity

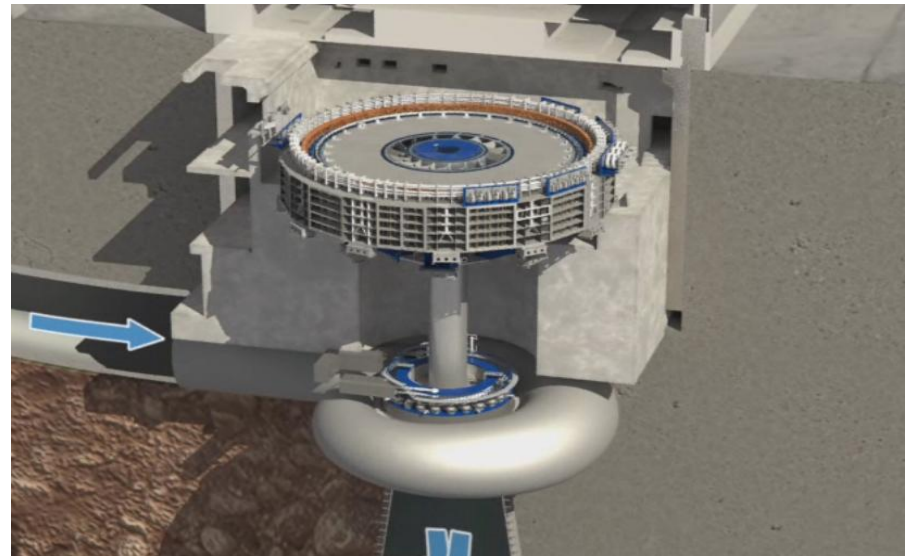
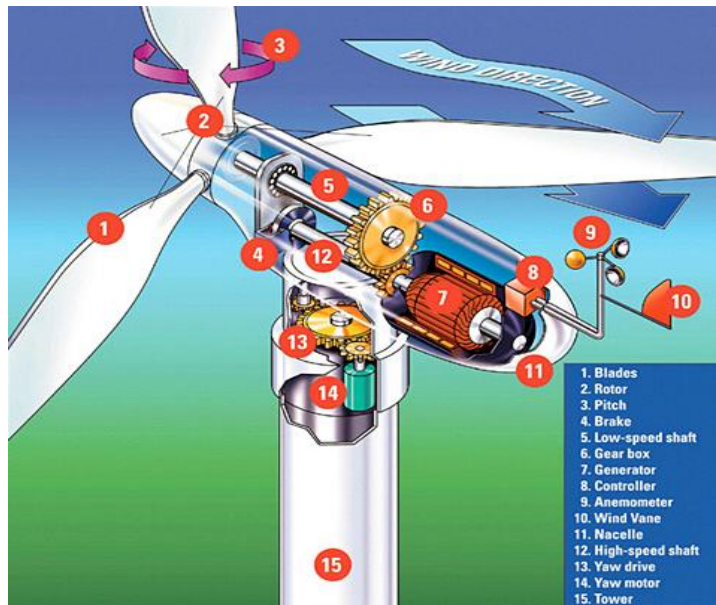
Gas/Steam turbines



Hydro turbines



Wind turbines

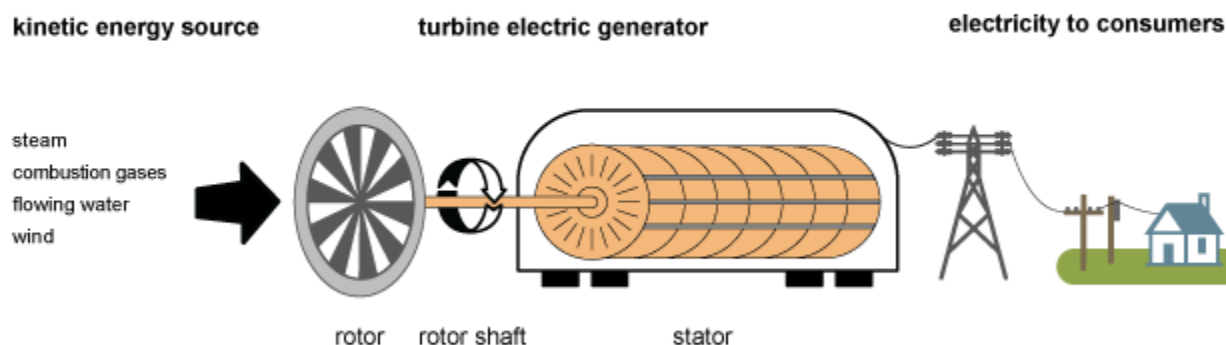


Electromechanical Energy Conversion

❖ Electromechanical Energy Conversion Devices

- Predominant way of generating and consuming electrical energy

Electricity generation from an electric turbine



Source: U.S. Energy Information Administration



❖ Electrical Energy Generation

- Almost all electricity produced is by **Synchronous Generators**

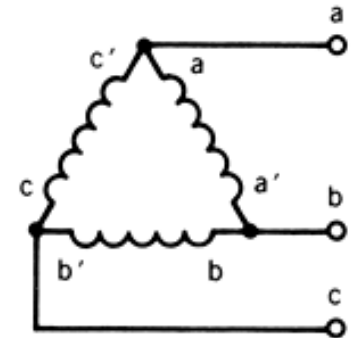
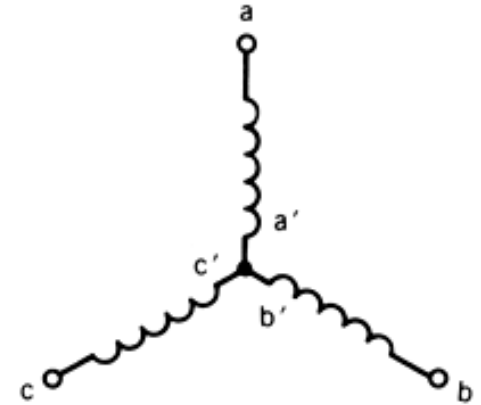
❖ Electrical Energy Utilization

- About 65% of all electricity is consumed by **motors: mostly induction**, less synchronous, etc.
- For industry it is about 85%



Stator Windings

- ✓ A typical stator would have 3-phases, which may be Y or Δ connected
- ✓ Windings may have multiple sections with parallel and/or series connection

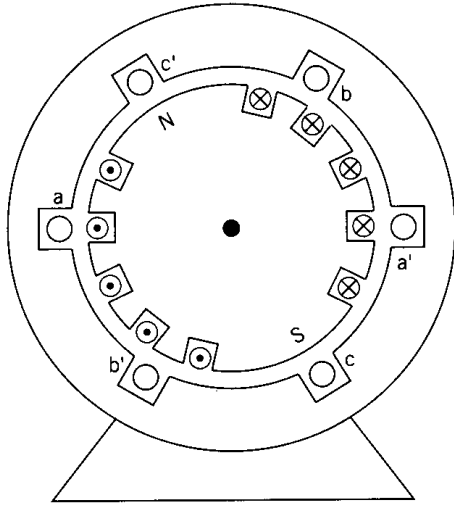


S. Ebrahimi EE 582, F-25, M-0

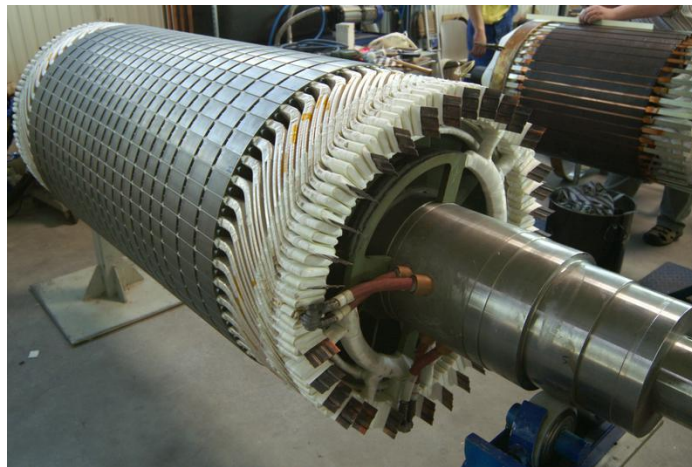
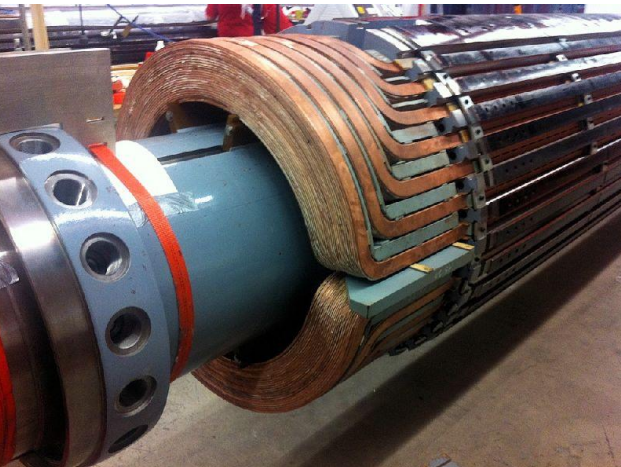
Synch. Mach. Rotor Types (Round)

✓ A typical round rotor has small number of magnetic poles (high speed)

2-Pole Round Rotor



4-Pole Round Rotor

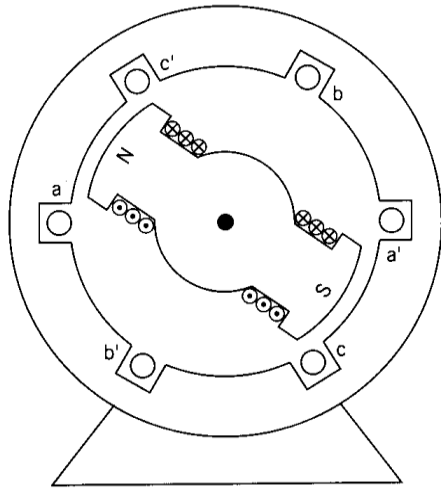


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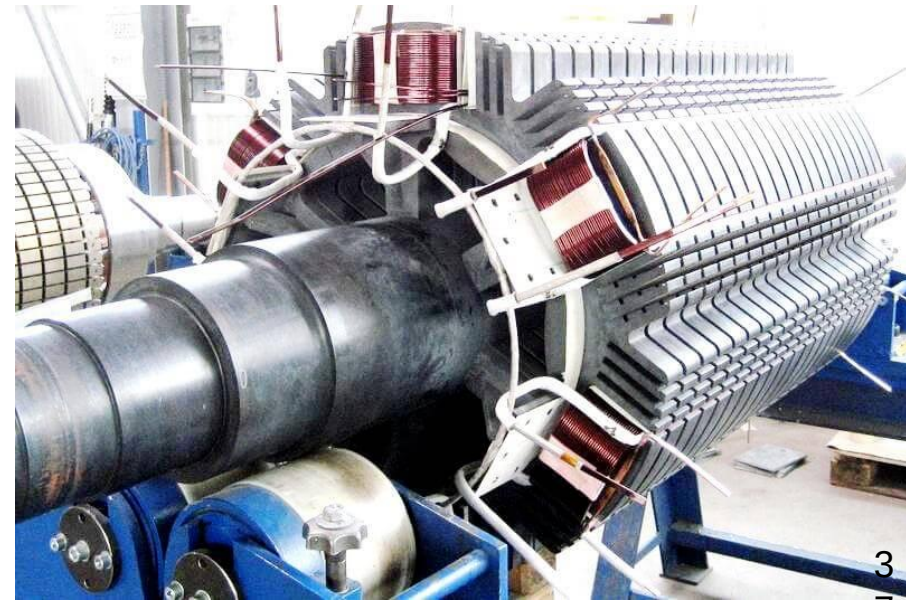
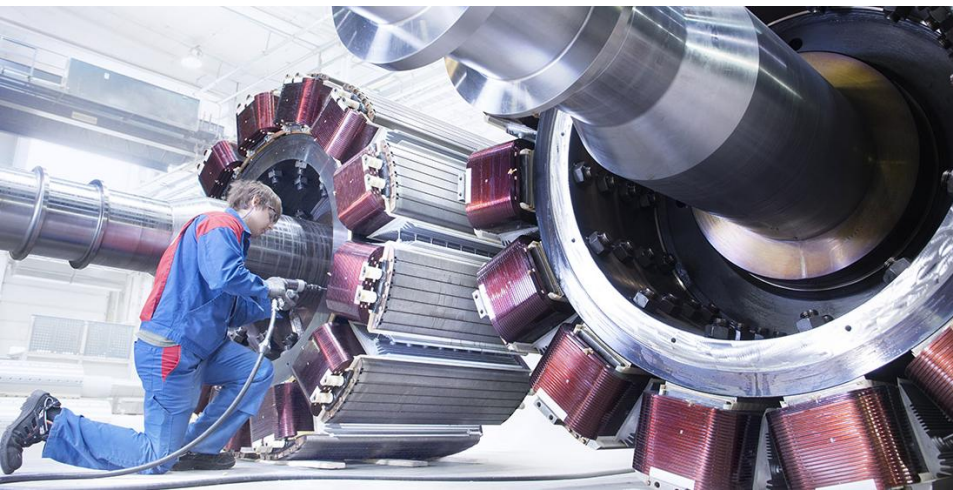
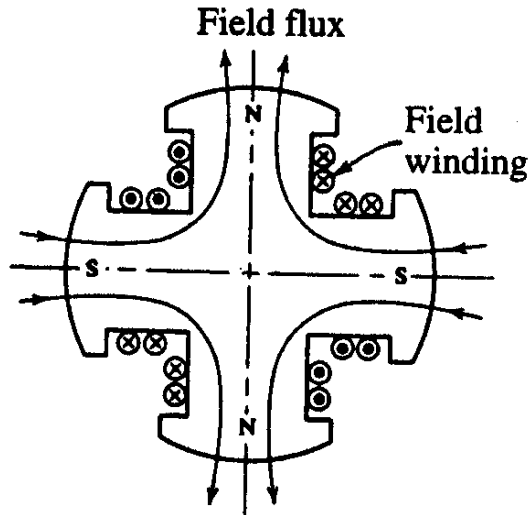
Synch. Mach. Rotor Types (Salient)

✓ A typical salient rotor may have more magnetic poles (lower speed)

2-Pole Salient Rotor



4-Pole Salient Rotor



Poles & Speed in Syn. Machines

7500 HP, 16kV Syn. Motor



8-Pole Salient Rotor

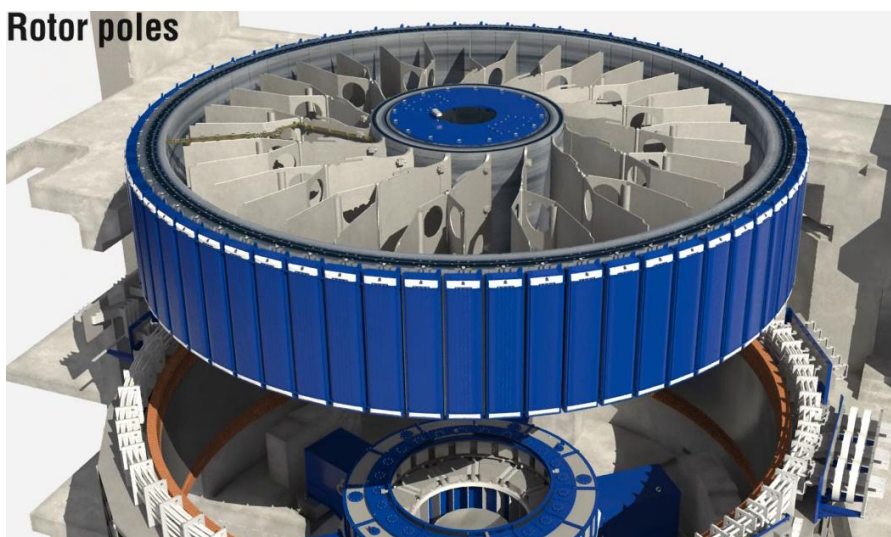


823MVA Hydro Generator, Brazil



64-Pole Salient Rotor Hydro Generator

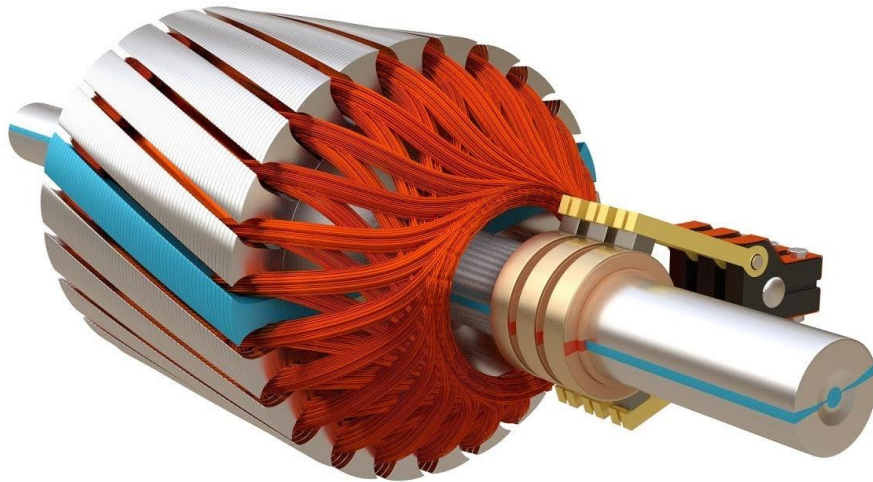
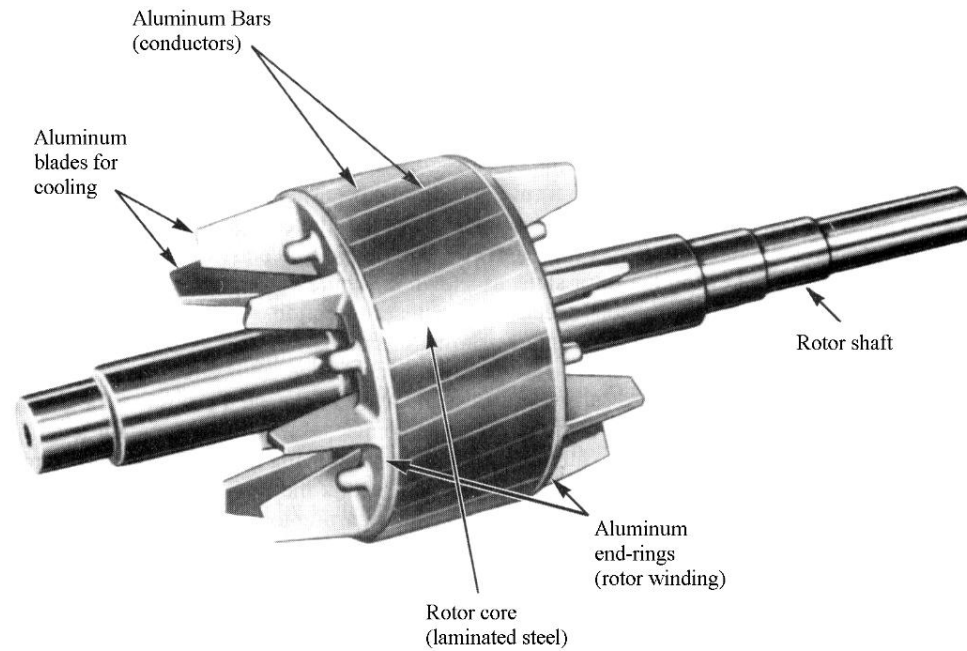
Rotor poles



Generation & Utilization of Electricity

- Use low number of poles (e.g. 2, 4, 8, etc.) for high-speed (e.g. 3600 – 900 rpm) low torque applications
 - Steam & thermal turbine generators, high speed motors, etc.
- Use high number of poles (e.g. 64..) for low-speed (e.g. 112.5 rpm) high torque applications
 - Hydro generators, direct-drive wind generators, low speed motors, etc.

IM Rotor Types

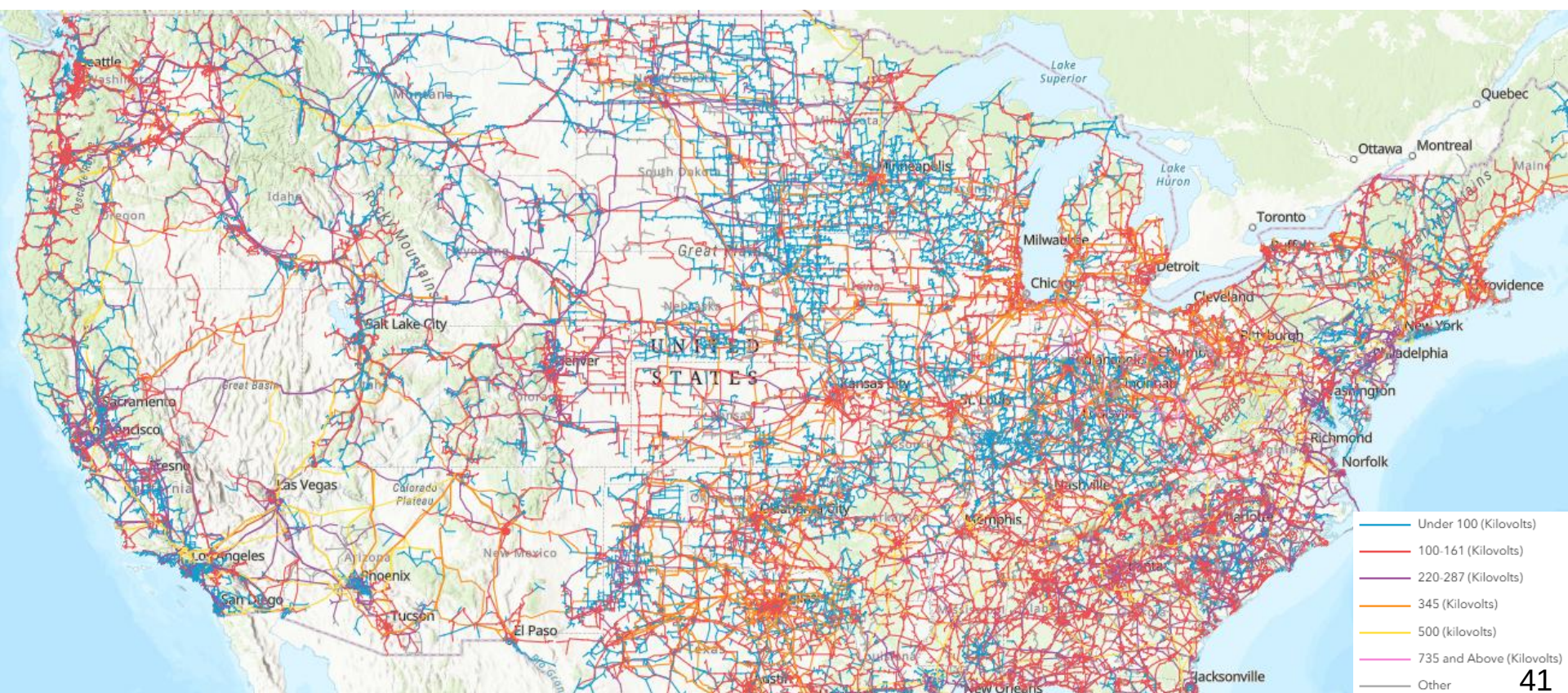


Variable Frequency Drives (VFDs)

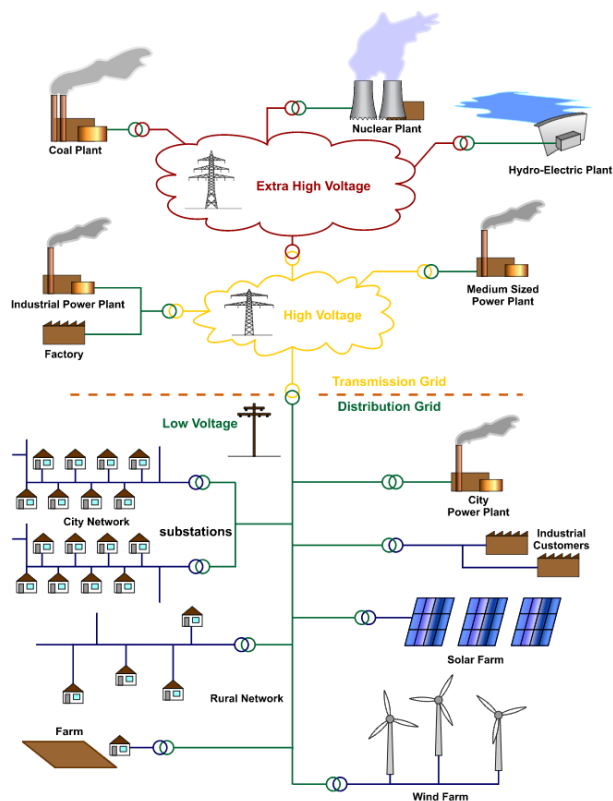


Large-Scale Network Simulation

- Power systems include many **energy conversion devices** (i.e., electric machines and power electronic converters)
- The behavior of such devices strongly influences system dynamics, stability, and efficiency
- For system studies (steady-state / transient / stability analysis), we mainly rely on **computer simulations** that are based on **mathematical models** of these devices



Large-Scale Network Simulation



➤ Require many offline and real-time simulations

- ✓ Design, palnning and analysis
- ✓ Monitoring and tuning

➤ Reliable simulations require models of components

- ✓ Accurate
- ✓ Fast and efficient



Major Topics Covered

- ✓ Introduction to software tools and methods for electromagnetic transient (EMT) simulations
- ✓ **Nodal-analysis-based methods for EMT simulations of power systems**
- ✓ **State-variable-based methods for EMT simulations**
- ✓ Space phasors and modeling of power systems in two-dimensional reference frames
- ✓ Control and modeling of voltage source converters (VSCs)
- ✓ EMT modeling and control of synchronous machines
- ✓ EMT modeling and control of induction machines
- ✓ Grid-following and grid-forming converters, their modeling and control
- ✓ Introduction to real-time simulations and constraints

Check the Canvas!

What to Know

Attendance: Try to attend all the lectures. Communicate in-advance if you need to be absent with a legitimate reason. No more than 3!

Office Hours: Email me for an appointment with “EE 582” in the subject line.

Class Notes: Trying to take notes is an effective way to learn!

Assignments: There will be 7–8 assignments throughout the course.

Simulation Project: Simulation of an IEEE Transaction paper related to the course topics.

Final Evaluation: Assignments (80%) – Project (20%)

Accommodations: Visit or call the Access Center

Academic Integrity: Don't cheat!

Score	Letter Grade
90 – 100	A
85 – 89	A–
80 – 84	B+
75 – 79	B
70 – 74	B–
below 70	Try not being here!!

Check the Canvas!